



**COMSATS University Islamabad,
Sahiwal Campus.**

**GlaucoVision : Multi-Model Detection and
Explanation of Glaucoma**

A Project Presented To

**COMSATS University Islamabad,
Sahiwal Campus**

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Computer Science (2022-2025)

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It is to certify that the final year project of BS(CS) “**GlaucoVision : Multi-Model Detection and Explanation of Glaucoma**” was developed by **Ali Murad (CIIT/SP22-BCS-023/SWL)**, **Ans Shahzad (CIIT/SP22-BCS-027/SWL)** under the supervision of “**Mr. Ali Sher Kashif**” and co supervisor “**Dr. Javed Ferzund**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Science.

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Executive Summary

GlaucoVision is an advanced diagnostic system designed to support early detection and clinical explanation of glaucoma using multiple AI-powered models. The project focuses on enhancing ophthalmic screening by providing automated analysis of fundus images, enabling both clinicians and patients to identify glaucoma indicators with improved accuracy.

The system integrates multi-model deep learning architectures, each specializing in optic disc detection, cup-to-disc ratio estimation, and glaucoma classification. By combining their predictions, GlaucoVision ensures reliable diagnostic support while minimizing false positives and false negatives. The platform further enhances medical understanding through an explanation module that highlights critical regions in retinal images, promoting transparency and trust in AI-assisted healthcare.

Designed with both medical professionals and patients in mind, GlaucoVision offers a clean, user-friendly interface where users can upload retinal images, view results, and access detailed visual explanations. Clinics and researchers benefit from an efficient workflow that automates key diagnostic steps and standardizes evaluation, making glaucoma screening more accessible.

This project follows a structured development lifecycle, including requirement analysis, system design, model training, and interface integration. Extensive evaluation—including accuracy testing, comparison between models, and visualization-based validation—ensures reliable performance. The system has undergone functional testing, integration testing, and user acceptance validation to ensure clinical usability and technical stability.

In summary, GlaucoVision provides a powerful and reliable AI-based solution for glaucoma detection and explanation. By combining deep learning models with a transparent explanation framework, the system enhances diagnostic confidence, supports early intervention, and contributes to improved eye-care outcomes. This innovative solution demonstrates how AI can play a transformative role in modern healthcare, offering accessible, accurate, and explainable glaucoma assessment for a wide range of users.

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Abbreviations

CUI	Comsats University Islamabad
SRS	Software Require Specification
SDD	Software Design Description
UC	Use Case
HCI	Human-Computer Interaction
DBMS	Database Management System
UT	Unit Testing
FT	Functional Testing
IT	Integration Testing
NoSQL	Not Only SQL
IDE	Integrated Development Environment
UI	User Interface
UX	User Experience

RGB	Red–Green–Blue (color space for images)
QA	Quality Assurance
PDF	Portable Document Format
LIME	Local Interpretable Model-Agnostic Explanations
FN	False Negative
FP	False Positive
CNN	Convolutional Neural Network
CDR	Cup-to-Disc Ratio
API	Application Programming Interface
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping

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Chapter 1

Introduction

1 Introduction

GlaucoVision is a transformative contribution to the field of ophthalmic healthcare, addressing the critical need for early and accurate detection of glaucoma—one of the leading causes of irreversible blindness worldwide. This specialized system has been meticulously designed to assist both clinicians and patients by offering an intelligent, automated, and explainable diagnostic platform centered around retinal fundus imaging.

Unlike traditional screening methods that rely solely on manual examination and specialist expertise, GlaucoVision utilizes advanced deep learning models to analyze optic nerve head structures, estimate key clinical markers, and provide reliable glaucoma predictions. This domain-specific approach ensures that each diagnostic output remains medically relevant and clinically actionable, bridging the gap between complex medical imaging and accessible healthcare support. Currently, early-stage glaucoma detection remains a challenge due to subtle structural changes that are often difficult to identify through conventional observation alone. The lack of accessible AI tools further widens this gap, especially in regions with limited specialist availability. GlaucoVision addresses this need by offering a comprehensive, accurate, and user-friendly system that enhances diagnostic capabilities while promoting early intervention.

The system's intuitive interface ensures that users—including ophthalmologists, trainees, and even non-specialist practitioners—can easily upload fundus images, view diagnostic results, and interpret AI-generated explanations. With clear visualizations that highlight critical regions of interest, GlaucoVision supports informed medical decision-making without requiring technical expertise.

In essence, GlaucoVision aims to improve glaucoma screening outcomes by providing an intelligent, explainable, and accessible diagnostic tool. This innovative platform enhances early detection efforts, empowers healthcare professionals, and contributes significantly to the future of AI-driven medical diagnostics.

students.

1.1 Brief Overview

GlaucoVision aims to develop an intelligent diagnostic system designed specifically to support the early detection and explanation of glaucoma using retinal fundus images. The system integrates multiple deep learning models to analyze optic nerve structures, estimate key clinical indicators such as the cup-to-disc ratio, and classify the likelihood of glaucoma with enhanced accuracy. By

combining these models, GlaucoVision offers a reliable, scalable, and medically meaningful approach to glaucoma screening.

GlaucoVision is a focused and domain-specific platform built to provide clinicians and patients with an accessible and interactive solution for eye-health evaluation. While many medical AI tools exist, GlaucoVision stands apart due to its specialized emphasis on glaucoma detection and explainability. It addresses the challenges faced in early diagnosis—particularly the subtle structural variations that are easily overlooked—by delivering clear visual explanations, highlighting critical regions, and supporting informed medical decisions. This dedicated diagnostic framework ensures that both accuracy and transparency remain central to the user experience. [1].

1.2 Relevance to Course Modules

The development of GlaucoVision directly applies and integrates various concepts and skills acquired throughout our Computer Science course.

i. Technical Skills:

This project required the use of programming languages, tools, and frameworks studied during the degree. Building the glaucoma detection model involved Python, machine learning libraries, and image-processing techniques - skills gained from our courses on Machine Learning and Artificial Intelligence. Similarly, developing the web interface and backend services applied concepts learned in Web Engineering and full-stack development. These skills were essential in shaping the core functionality of the entire system.

ii. Software Engineering Principles:

The project follows the principles of software engineering, starting from requirement gathering to system design, development, testing, and deployment. Every stage of GlaucoVision was carried out using structured development practices to ensure that the system remains easy to maintain, scalable, and reliable. Concepts from software engineering, software architecture, and quality assurance helped ensure that the system was built in a professional and organized way.

iii. Database Design:

A strong understanding of database concepts was required to store patient records, diagnostic results, and medical reports. Knowledge gained from Database Management Systems helped us design a secure and efficient data storage structure. Proper indexing and data organization ensured that medical records load quickly and can be retrieved whenever the clinician needs them.

iv. System Architecture:

Designing the system required an understanding of modern web application architecture. Concepts from Parallel and Distributed Systems were applied to integrate the AI model with the backend, manage data flow, and create a structured and scalable platform. This allowed the system to handle image uploads, run predictions, store results, and display them in an organized way for clinicians.

v. UI/UX Design:

A user-friendly interface is essential for doctors who use the system. Principles learned from Human-Computer Interaction (HCI) helped us design a clean, simple, and intuitive dashboard. The layout of the upload screen, prediction results, recent scans, and reports was created to make navigation easy for clinical staff, ensuring that even users with minimal technical background can use the system comfortably.

vi. Project Management:

The planning, execution, and coordination of the project were guided by our project management concepts. Using Agile methodology allowed us to break development into smaller, manageable stages, improving the system step by step. Regular planning, progress tracking, and task division helped in the smooth completion of the project within the given timeline.

vii. Networking

Since the system uses real-time communication between the web interface, backend server, and AI model, networking concepts played an important role. Knowledge of HTTP requests, APIs, and server communication enabled us to ensure smooth interaction between all components. This allowed clinicians to upload images, receive predictions, and download reports without delays.

1.3 Project Background

The need for the GlaucoVision arises from several major challenges faced by doctors and healthcare staff when diagnosing glaucoma, especially in settings where resources are limited. Glaucoma is a disease that develops gradually and often shows no clear symptoms in its early stages, which makes early detection extremely difficult. Many clinics depend entirely on manual examination of fundus images, and this process requires trained specialists, time, and consistent accuracy, which is not always available.

Before GlaucoVision, there was no dedicated system where clinicians could upload fundus images and instantly receive AI-based support for glaucoma detection. Doctors often had to rely on

traditional methods, which were sometimes slow and did not always help in identifying subtle signs of early-stage glaucoma. Existing medical tools were either too expensive, difficult to use, or lacked explainability, meaning clinicians could not clearly understand how an AI decision was made.

Additionally, patient information and diagnostic histories were usually stored manually or scattered across different systems. This made it hard for doctors to track changes over time or maintain a structured clinical record. Without a centralized and intelligent platform, healthcare professionals faced difficulty in managing patient scans and accessing previous results when needed.

To overcome these challenges, GlaucoVision was developed as a dedicated clinical platform that allows doctors to:

- Upload fundus images for automated glaucoma detection.
- View explainable AI results that highlight important image regions.
- Access and manage patient diagnostic histories through a unified dashboard.

By addressing these limitations, GlaucoVision improves early detection, makes the diagnostic process more organized, and supports clinicians in making informed and timely decisions.

1.4 Literature Review

Table 1.4.1: Literature Review

Application Name	Weakness	Proposed Project Solution (CUI Sphere)
Deep Glaucoma	Uses only one imaging modality; limited diagnostic depth; lacks explainability features.	Enhanced fundus-based deep learning with explainability maps that highlight key image regions.
Orbis Cybersight AI	Designed mainly for remote assistance; does not include record management or complete clinical workflow.	Provides a unified clinical dashboard with patient history, scan tracking, AI predictions, and report generation.

Traditional Retinal Image Analysis Systems	Lower accuracy in early-stage detection; limited visualization support.	Utilizes refined deep learning techniques with visual heatmaps to support early detection and assist clinicians in understanding model decisions.
Manual Hospital Record Systems	Time-consuming, unstructured, prone to errors, and lacks AI-based diagnostic support	Offers centralized patient records, automated predictions, downloadable reports, and an organized digital workflow tailored for clinics.

1.5 Analysis from Literature Review

The analysis of existing glaucoma-related diagnostic systems highlights key gaps in automation, explainability, and clinical workflow integration - areas GlaucoVision is specifically designed to address. Below is a breakdown of the systems reviewed and how GlaucoVision improves upon their limitations:

1.5.1 Deep Glaucoma

Many AI systems like Deep Glaucoma rely mainly on OCT or single-modality images, which limits the reliability of early detection. They also do not offer clear visual explanations of their predictions, making it difficult for clinicians to trust the decision-making process. GlaucoVision addresses this by focusing on improved fundus-based analysis along with explainability heatmaps that highlight the key areas the model used to reach its conclusion, giving doctors better confidence and clarity. ecosystem.

1.5.2 Orbis Cybersight AI

Although Orbis Cybersight AI is helpful for remote screening, it does not integrate with hospital workflows or support complete patient record management. Doctors still need separate systems for storing scans, tracking history, and generating reports. GlaucoVision fills this gap by offering a structured clinical dashboard where practitioners can upload images, view predictions, access records, and manage patient information - all within a single platform. relevance.

1.5.3 Retinal Image Analysis Tools

Traditional fundus-only analysis tools often struggle with early detection due to limited depth and lack of detailed interpretability. They generally do not provide strong visual indicators for

clinicians. GlaucoVision improves this by applying enhanced deep learning techniques and generating explainable visual overlays that show exactly which parts of the retinal image contribute to a positive or negative glaucoma prediction.

1.5.4 Manual Clinical & Hospital Record Systems

Most hospitals still use manual or partially digitized systems for storing patient data, which can be slow, unorganized, and prone to errors. These systems also lack AI support for diagnosis, forcing doctors to rely entirely on visual judgment. GlaucoVision transforms this by offering a centralized, digital record system combined with AI predictions, downloadable reports, and a user-friendly dashboard designed for clinical efficiency.

1.6 Methodology and Software Lifecycle

GlaucoVision is a software project that involves multiple components, including machine learning, image processing, a clinical dashboard, and patient record management. Since these features require continuous refinement and improvements over time, it is important to follow a development methodology that is flexible, collaborative, and able to adapt to changing requirements. For this reason, the Agile methodology was selected, as it supports iterative development and consistent enhancement throughout the project.

1.6.1 Rationale behind Selected Agile Model

Agile is a development approach that focuses on flexibility, teamwork, and ongoing improvement. It is especially suitable for projects like GlaucoVision, where model accuracy, system usability, and clinical workflow integration must be continuously refined. As the system depends on real medical use cases, feedback from clinicians, supervisors, and test users plays an important role in shaping the final outcome.

Agile allows the development team to build the system in smaller parts, test each part, and improve it based on feedback. This approach works well for AI-based projects because model performance, interface clarity, and backend integration often need repeated adjustments.

Agile is based on the following principles:

- **Iterative development:** Development is divided into small cycles, where new features—such as prediction modules, explainability reports, or dashboard sections—are added or improved in each iteration.
- **Incremental delivery:** Instead of completing everything at once, the system is delivered in working increments. Early versions focused on basic image uploads and prediction, while later increments added reports, patient history, and analytics.

- **Continuous feedback:** Feedback from peers, supervisors, and potential users helps identify improvements in model accuracy, usability, and system workflow.
- **Adaptive planning:** Development plans are adjusted based on challenges that arise during model training, interface design, or backend integration, ensuring the system evolves smoothly.

1.7 Summary

This chapter discussed the purpose of the project, its connection to major course concepts, and insights gained from the literature review. It also explained the development methodology used for GlaucoVision and highlighted why Agile is the best fit for a system that requires continuous refinement, user feedback, and iterative improvement.

Chapter 2

Problem Definition

2 Problem Definition

This chapter discusses the specific problem that GlaucoVision aims to solve. It highlights the challenges present in the current healthcare workflow and explains why a dedicated system is needed. It also outlines the intended outcome of the proposed solution.

2.1 Problem Statement

Glaucoma remains one of the major causes of irreversible blindness, yet early detection continues to be a significant challenge in many healthcare settings. Most clinics rely on manual examination of fundus images, which requires experienced ophthalmologists and specialized tools. As a result, many patients are diagnosed only when the disease has already progressed, making treatment less effective and increasing the risk of permanent vision loss.

Moreover, there is no centralized or automated digital system that allows clinicians to upload fundus images, receive instant AI-based predictions, or access clear visual explanations to support diagnosis. Doctors often work with scattered records, manual report writing, and fragmented tools, which makes it difficult to track patient histories or compare previous scans. Without an organized platform, the diagnostic workflow becomes slow, time-consuming, and prone to inconsistencies.

To address these challenges, GlaucoVision is being developed as a unified clinical support system that streamlines glaucoma detection and patient record management. The system enables clinicians to upload fundus images and instantly receive AI-generated predictions along with visual explanation maps. Additionally, GlaucoVision provides a structured patient dashboard where doctors can view past scans, save diagnostic reports, and manage patient histories in one place.

The absence of such an integrated and intelligent diagnostic platform highlights the need for GlaucoVision, ensuring that clinicians can detect glaucoma earlier, work more efficiently, and provide better patient care.

2.2 Deliverables and Development Requirements

The GlaucoVision project includes several key deliverables and requirements to ensure that the platform works smoothly for both clinicians and patients. Since both user types have access to the system's major features, the deliverables focus on creating a unified and easy-to-use experience for everyone. These include:

- **User Interface (UI) Design:** Develop a clean and user-friendly interface that allows both clinicians and patients to navigate the system easily. The interface should clearly present uploaded images, prediction results, reports, and personal records in a simple and organized layout.
- **User Registration & Profiles:** Implement a secure login and account system where both clinicians and patients can create profiles. Each user can view and manage their own information, access reports, and interact with system features without restrictions.
- **Fundus Image Upload Module:** Provide an upload feature that allows any user - clinician or patient - to submit fundus images for glaucoma analysis. The system should support standard image formats and ensure fast and smooth uploading.
- **AI Prediction and Explanation Reports:** Integrate an AI model that analyzes uploaded fundus images and provides glaucoma predictions. Along with the result, the system will generate explainability maps that highlight key areas of the image, allowing all users to understand the reasoning behind the prediction.
- **Centralized Records and Reports:** Create a unified record system where both user types can view previous uploads, predictions, and reports. This ensures that all diagnostic information stays organized and easily accessible whenever needed.
- **Dashboard for Users:** Develop a shared dashboard where both clinicians and patients can access the same system features. This includes recent scans, prediction summaries, reports, and personal history - all displayed in a simple, easy-to-navigate format.
- **Security and Privacy:** Ensure strong security for all user accounts by using encrypted storage, secure authentication, and proper access controls. Even though both user types can access all features, their data must remain safe and confidential.
- **Scalability:** Design the system to support a growing number of users, image uploads, reports, and records without affecting performance or speed.
- **Backend Integration:** Develop a reliable backend that connects the web interface, AI model, and database smoothly. This ensures fast communication for uploads, predictions, and record retrieval.
- **Documentation:** Prepare clear user guides and documentation to help both clinicians and patients understand system features, upload images, and interpret reports.
- **Testing and Quality Assurance:** Perform thorough testing including functional testing, user acceptance testing, and model evaluation - to ensure that every feature works correctly for all users.

- **Support and Guidance:** Provide simple instructions, help panels, or built-in hints so users can easily understand how to use each part of the platform.
- **Regular Updates and Maintenance:** Plan for ongoing improvements based on feedback, including model updates, new features, performance enhancements, and security upgrades.

These deliverables and requirements ensure that GlaucoVision becomes an accessible, accurate, and user-friendly glaucoma detection system for both clinicians and patients.

2.3 Current System

Currently, most healthcare facilities do not have a centralized platform for detecting glaucoma or managing patient eye-health records in an organized way. Doctors usually rely on manual examination of fundus images, which requires expertise, time, and careful attention. Because this process is completely manual, it can lead to delays, inconsistent results, and difficulty in identifying early signs of glaucoma.

In many clinics, patient reports, previous scans, and medical histories are stored separately or handled on paper, making it challenging to retrieve past information quickly. This lack of structure often creates gaps in diagnosis, as doctors cannot easily compare current images with previous ones. Patients also do not have a simple way to access their reports or understand their own eye-health status.

Additionally, there is no intelligent system that can automatically analyze fundus images or generate visual explanations to support clinical judgment. Without such tools, both clinicians and patients may miss important details that could help in early detection and treatment.

The absence of a unified, AI-assisted, and easy-to-use diagnostic platform highlights the need for GlaucoVision - ensuring that scans can be analyzed quickly, reports can be stored properly, and both patients and clinicians can view everything in one organized system.

Chapter 3

Requirement Analysis

3 Requirement Analysis

3.1 Use Case Diagrams

In this Use case diagrams, we show the behavior of our system. This enables you to visualize the different types of roles in a system and how those roles interact with the system.



Figure 3.1: GlaucoVision Use Case Diagram

3.2 Use Case Descriptions

The table below indicate comprehensive use case descriptions

Table 3.2.2: Use case description for Sign Up

Use case ID	UC-01
Use case name	Sign Up
Actors	Clinician, Patient
Description	A clinician or patient creates a new account to access the system.
Trigger	User selects the “Sign-up” option.
Pre-condition	User must have a valid email.
Postcondition	A new account is created and saved in the system.
Normal flow	User opens the website. Clicks “Sign-up”. Enters required details. System verifies information. Account is created successfully.
Alternative flow	User enters an already registered email The system notifies the user that the email is already registered. The system suggests using the password recovery option.
Exceptions	System down - user asked to try again later.
Business rules	Password must meet security requirements.

Table 3.2.3: Use case description for Log-In

Use case id	UC-02
Use case name	Log-in
Actors	Clinician, Patient
Description	User logs into the system using their credentials.
Trigger	User selects “Log-in”.
Pre-condition	User must already be registered.
Postcondition	User gains access to dashboard and system features.
Normal flow	User enters email and password. System verifies credentials.

	User is logged in successfully.
Alternative flow	The Incorrect credentials entered The system notifies the clinician of incorrect details. The clinician or patient is prompted to re-enter credentials.
Exceptions	The server is down. The system notifies the clinician or patient and suggests trying later.

Table 3.2.4: Use case description for Upload Fundus Image

Use case id	UC-03
Use case name	Upload Fundus Image
Actors	Clinician, Patient
Description	User uploads a fundus image for analysis.
Trigger	User clicks “Upload Image”.
Pre-condition	User must be logged in.
Postcondition	Image is stored and ready for processing.
Normal flow	User selects image file. System checks format and size. Image is uploaded successfully.
Exceptions	The server is down. Cannot upload fails.

Table 3.2.5: Use case description for Process & Display Diagnosis Result

Use case id	UC-04
Use case name	Process & Display Diagnosis Result
Actors	Clinician, Patient
Description	System analyzes the uploaded image and displays the glaucoma prediction.
Trigger	User clicks “Analyze”.
Pre-condition	Valid fundus image must be uploaded.
Postcondition	Prediction result appears on screen.
Normal flow	System sends image to ML model. Model processes image. System displays diagnosis result.

Table 3.2.6: Use case description for Generate Explainability Report

Use case id	UC-05
Use case name	Generate Explainability Report
Actor	ML Model
Description	System generates heatmap-based explanation for the prediction.
Trigger	User requests “Explainability Report”.
Precondition	Diagnosis must be completed.
Postcondition	Explainability map and text summary are displayed.
Alternative flow	Processing delay - user notified.
Normal flow	System processes image. Generates heatmap. Displays explainability report.

Table 3.2.7: Use case description for Download Explainability Report

Use case id	UC-06
Use case name	Download Explainability Report
Actor	Clinician, Patient
Description	User downloads the generated AI report.
Trigger	User clicks “Download Report”.
Precondition	Report must already be generated.
Postcondition	File is downloaded successfully.
Alternative flow	Report is missing. System shows message.
Normal flow	User clicks download. System prepares file. File is saved to user’s device.

Business Rule	Reports should be available for download for at least 30 days.
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Table 3.2.8: Use case description for Access Dashboard

Use case id	UC-07
Use case name	Access Dashboard
Actors	Clinician, Patient
Description	User accesses the main dashboard containing all features.
Trigger	User clicks “Dashboard”.
Precondition	User must be logged into their account.
Postcondition	Dashboard loads successfully.
Normal flow	System retrieves user data. Displays dashboard.
Exceptions	Server down. Database unavailable.

Table 3.2.9: Use case description for Chatbot

Use case id	UC-08
Use case name	Chatbot
Actors	Clinician, Patient
Description	User interacts with AI chatbot for assistance.
Trigger	User clicks “Chatbot”.
Pre-condition	User is logged in.
Postcondition	Chat response is displayed.
Normal flow	The user opens the chatbot from the dashboard. The user types a query. The chatbot processes the query. The chatbot generates a response. The user continues interacting or closes the chatbot window.

Table 3.2.10: Use case description for Recent Scans

Use case id	UC-09
Use case name	Recent Scans
Actors	Clinician, Patient
Description	User views recently uploaded fundus images and results.
Trigger	User selects “Recent Scans”.
Pre-condition	User must have history data.
Postcondition	Recent scans are displayed.
Normal flow	User opens the dashboard. User clicks “Recent Scans.” The system retrieves all recent scan records. The system displays thumbnails, prediction results, and dates. User can select any scan to view full details or download report.

Table 3.2.11: Use case description for Billing

Use case id	UC-10
Use case name	Billing
Actors	Clinician
Description	Clinician views subscription or usage billing details.
Trigger	Clinician clicks “Billing”.
Pre-condition	Billing module active.
Postcondition	Billing information displayed.
Normal flow	Clinician opens dashboard. Clinician selects “Billing.” System loads subscription plan and payment history. Clinician reviews details.
Exceptions	Payment Gateway Unavailable

Table 3.2.12: Use case description for Account Setting

Use case id	UC-11
Use case name	Account Setting
Actors	Clinician, Patient
Description	User updates personal details, password, or preferences.

Trigger	User selects “Account Setting”.
Pre-condition	User logged in.
Postcondition	Updated information saved.
Normal flow	User opens “Account Settings.” User updates profile details (name, password, picture). System validates the entries. System saves changes in the database. System confirms successful update.
Exceptions	The error occurs in the application. Changes are not saved; system alerts user.

3.3 Functional Requirements

Functional requirements are crucial in software development because they clearly define what a system should do to meet the needs of its users. These requirements describe the specific functions, features, and behavior the system must exhibit, such as user authentication, data processing, or content display. Their importance lies in providing a clear foundation for both developers and stakeholders, ensuring everyone understands the system's expected capabilities. By outlining these functions early in the development process, functional requirements help prevent miscommunication, reduce the risk of costly changes later, and allow for better planning, testing, and validation. Ultimately, they guide the development process and ensure the final product aligns with user expectations and project goals [4].

The functional requirements of our project focus on creating a complete glaucoma detection platform that supports both clinicians and patients. These requirements include secure user access, fundus image upload, AI-based diagnosis, explainability reports, an organized dashboard, and features that help users view their diagnostic history and download reports. Each requirement plays an important role in ensuring that GlaucoVision works smoothly and provides an accurate, user-friendly experience for everyone involved. These requirements include:

3.3.1 User Authentication and Authorization

The system must provide secure login and registration for all users. Both clinicians and patients should be able to create accounts, log in securely, and manage their personal information. The system should ensure that only verified users can access the platform.

3.3.2 Fundus Image Upload and Preprocessing

Users must be able to upload fundus images for analysis. The system should accept valid image formats, perform basic preprocessing (such as resizing or quality checks), and prepare the image for AI-based prediction.

3.3.3 AI-Based Glaucoma Detection

The system should analyze uploaded fundus images using a trained AI model. After processing, it must display a clear result indicating whether glaucoma is detected or not. The diagnosis should also include a confidence score for better clarity.

3.3.4 Explainability Report Generations

The system must generate an explainability report that shows highlighted regions on the fundus image to help users understand how the model reached its decision. This report should include a heatmap and a brief text summary.

3.3.5 Downloadable Reports

After analysis, users should be able to download a detailed diagnostic report. This report should contain the prediction result, confidence score, explainability heatmap, and basic patient information.

3.3.6 User Dashboard

The platform must include a dashboard where both clinicians and patients can view recent scans, prediction results, uploaded images, and reports. The dashboard should present information clearly and help users navigate through different features easily.

3.3.7 Recent Scans and History Tracking

The system should maintain a history of previously uploaded fundus images and their results. Users must be able to view these past entries and compare older reports with current ones whenever needed.

3.3.8 Chatbot Assistance

A simple AI-powered chatbot must be available to help users with basic questions, system navigation, and general guidance. The chatbot should respond quickly and provide helpful suggestions when a query is not understood.

3.3.9 Account Settings Management

Users must be able to update their personal details, change passwords, and manage general preferences through the account settings section. The system should validate all updates before saving them.

3.3.10 Security and Privacy Controls

The system must protect all medical data using encrypted storage, secure authentication, and access control. Patient-related information should remain confidential and accessible only to authorized users.

3.4 Nonfunctional Requirements

Non-functional requirements are those that establish the parameters used to evaluate how well a system performs. Instead of describing specific features, these requirements define the limitations, quality standards, and conditions under which the system must operate.

3.4.1 Performance

The system should process fundus images quickly and generate predictions without long delays. Uploading images, viewing results, and generating explainability reports must feel smooth and responsive. be responsive and provide quick loading times, even during peak usage periods.

3.4.2 Usability

The platform should be easy for both clinicians and patients to use. The interface must be clean, simple to navigate, and designed so users can easily upload images, check diagnosis results, and access reports.

3.4.3 Security

Since the platform deals with medical data, all information must be protected. This includes secure login, encrypted storage, and strict access control to ensure patient records remain private.

3.4.4 Availability

This system must be always accessible, 24 hours a day, seven days a week. It should be available whenever users need it.

3.4.5 Maintainability

The system should be easy to maintain and update. Developers must be able to fix bugs, improve features, and update the AI model without disrupting existing functionality.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

The system architecture describes how the glaucoma detection web application functions and explains how different components interact to perform diagnosis, generate reports, and manage clinical information.

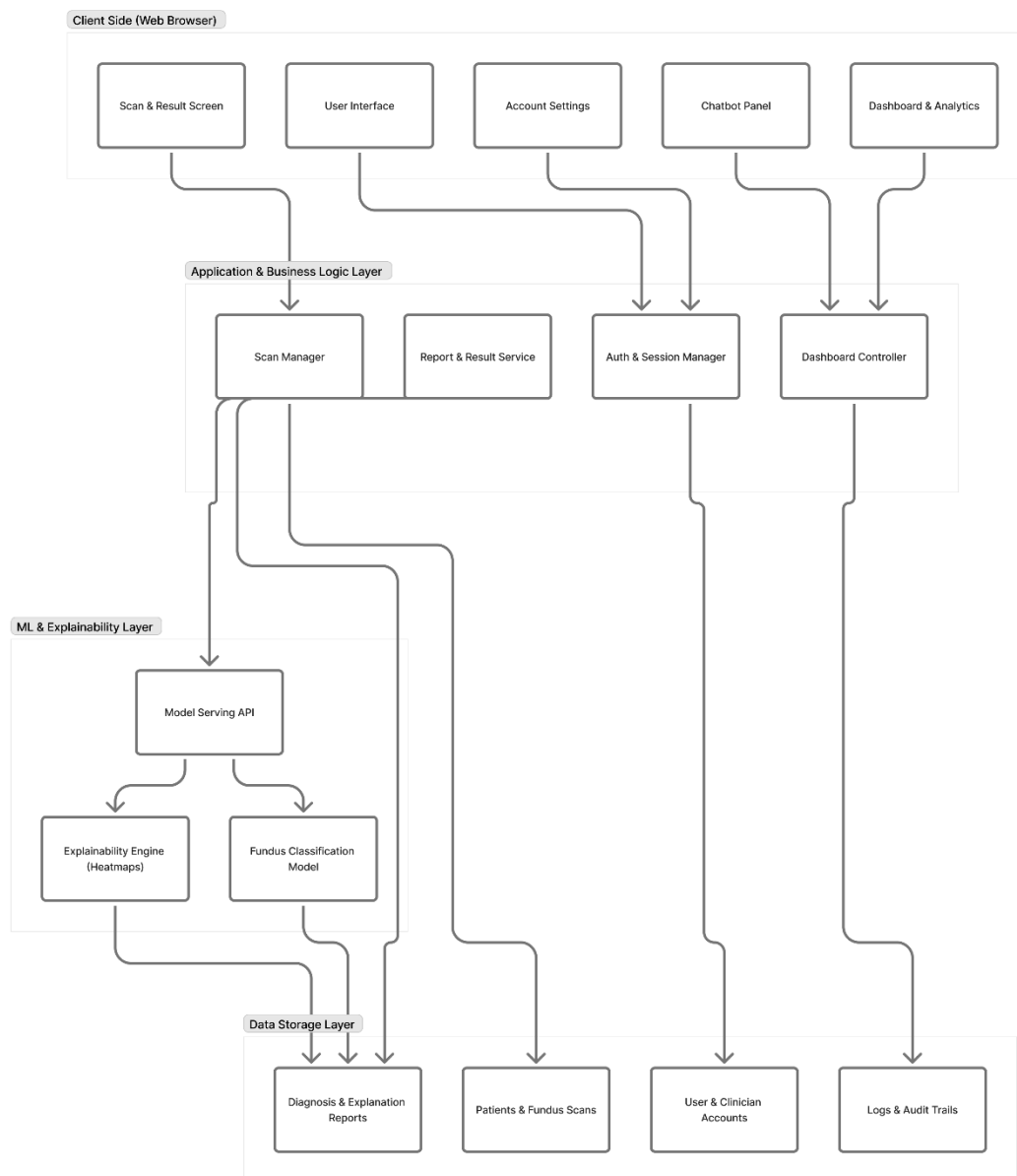


Figure 4.1: Glaucoma Web App Architecture

The architecture of the Glaucoma Detection Web App follows a client-server model with a layered and modular approach to ensure smooth processing, maintainability, and an efficient user experience. The client side runs through a web browser and contains all major screens such as Login, Signup, Upload Scan, Diagnosis Result, Dashboard, Recent Scans, Chatbot, and Account Settings, all connected through a unified and responsive user interface. The business logic layer handles essential system operations including user authentication, processing uploaded fundus images, generating diagnosis and explainability reports, and preparing data for dashboard analytics. This separation keeps the system logic independent from the interface and ensures clean communication between layers.

The backend integrates the machine learning model for glaucoma detection and the explainability module responsible for generating heatmaps, allowing the system to provide transparent AI-based predictions. A structured data storage layer maintains user accounts, patient details, uploaded fundus images, diagnosis results, and related analytics, enabling efficient retrieval whenever needed. The overall architecture supports smooth interaction between components, making the system reliable, easy to navigate, and capable of future enhancements.

4.2 Data Representation

In the Glaucoma Detection Web App, data is stored in a structured and organized manner to ensure efficient retrieval of user, patient, and diagnostic information. Unlike unstructured storage approaches, the system uses clearly defined data schemas to store fundus images, diagnosis results, explanation heatmaps, and patient details in an easily accessible format. This structured representation enables seamless data flow between the client interface, backend logic, and machine learning components, ensuring that clinicians can quickly access past scans, review reports, and track patient progress.

By maintaining well-defined data formats for uploads, predictions, and reports, the system ensures that information is encoded and decoded reliably, resulting in faster loading times, accurate record retrieval, and an overall optimized user experience. The organized schema allows for scalable expansion as new features, models, or diagnostic parameters are introduced. Role-based access control can be efficiently implemented to ensure data security and patient privacy. Furthermore, structured datasets facilitate easier analytics, auditing, and performance monitoring, enabling continuous improvement of the glaucoma detection system.

4.3 Process Flow

The Process Flow Representation illustrates the sequence of steps involved in using the Glaucoma Detection Web App. It provides a clear overview of how users interact with the system to upload scans, view diagnosis results, and manage their accounts.

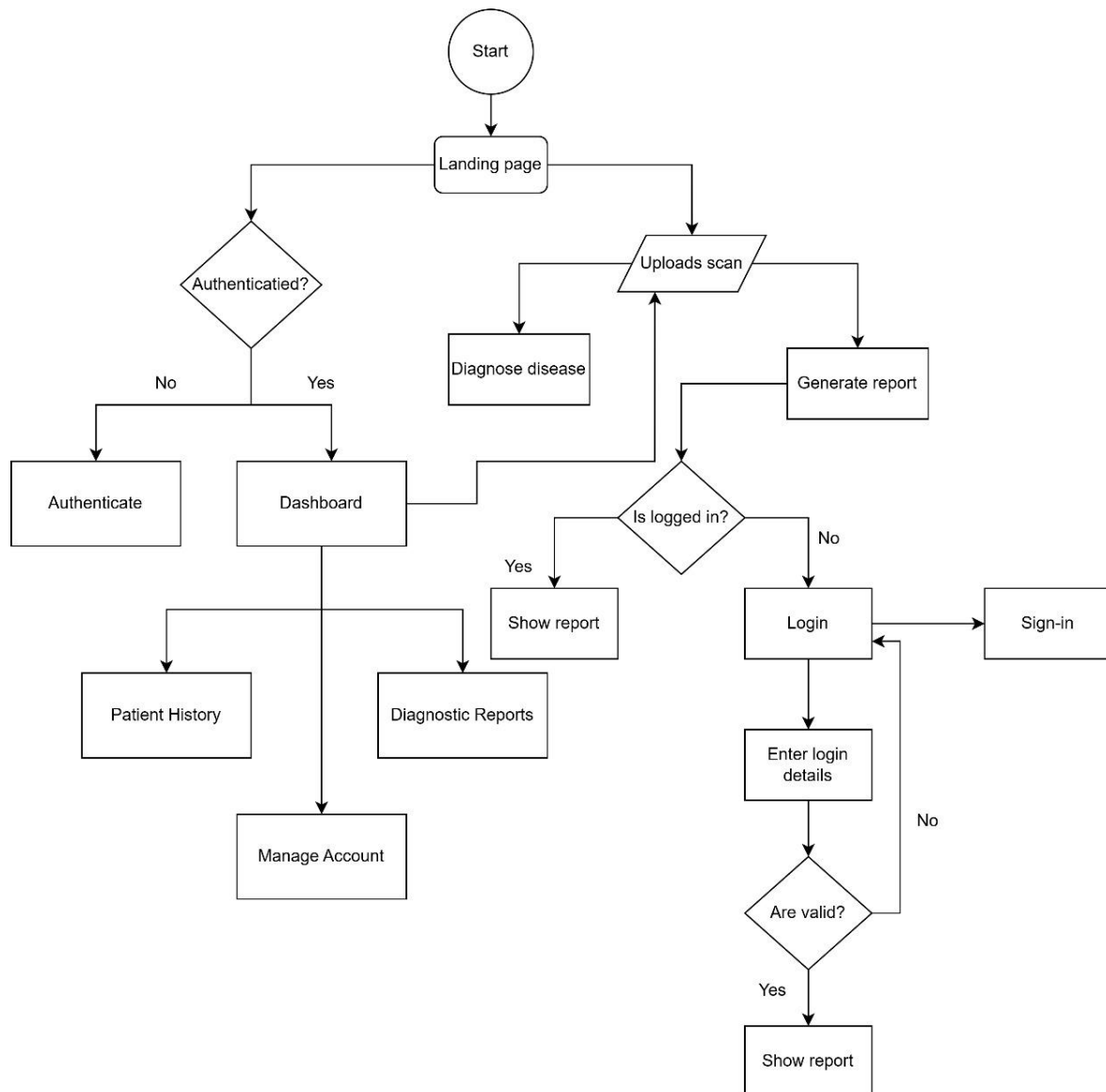


Figure 4.2: Glauco Vision Process Flow

4.3.1. Start and Landing Page

The process begins at the Landing Page, which serves as the entry point for all users. From here, the system determines whether the user is authenticated and what actions they want to perform, such as uploading a scan or accessing existing reports.

4.3.2. Authentication Check

Upon arriving at the Landing Page:

- If the user is already authenticated, they are directed to the Dashboard.
- If not, they must go through the authentication process, which may include entering login credentials or signing in through a standard login interface.

4.3.3. Dashboard Functionalities

Authenticated users gain access to the Dashboard, which provides key system features, including:

- **Diagnostic Reports** accessing detailed glaucoma analysis results
- **Recent Scans** quickly reviewing the most recently uploaded images and their outcomes
- **Analytics** viewing basic statistical insights related to system usage and patient scans
- **Chatbot** interacting with a simple AI-based chatbot for assistance.
- **Manage Account** updating personal information and account settings

This screen serves as the central hub for navigation inside the system.

4.3.4. Scan Upload and Disease Diagnosis

From the Landing Page, users can upload a fundus scan to initiate the diagnostic process. Once an image is uploaded:

- The system processes the scan using the AI model designed for glaucoma detection
- The model generates predictions along with explanation outputs (e.g., heatmaps)
This step allows the user to obtain an automated analysis of the retinal scan.

4.3.5. Report Generation and Access

After the disease diagnosis is complete:

- The system generates a diagnostic report containing prediction results and explanation visualizations
- If the user is already logged in, the report is displayed immediately
- If not, the system redirects the user to the Login interface before granting access to the report

4.3.6. Login Flow for Report Access

If an unauthenticated user attempts to view a report:

- They must enter valid login credentials
- If the credentials are valid, the system grants access and displays the report
- If the credentials are incorrect, the user is prompted to retry until successful

4.4 Sequence Diagram

The following sequence diagrams illustrate the operational flow of the Glaucoma Detection Web App. Each diagram focuses on a specific phase of the system, showing how users interact with the application and how backend components respond during key processes.

Phase 1: Authentication

This sequence diagram explains how a clinician or patient logs into the system. The user begins by entering their login credentials on the web interface.

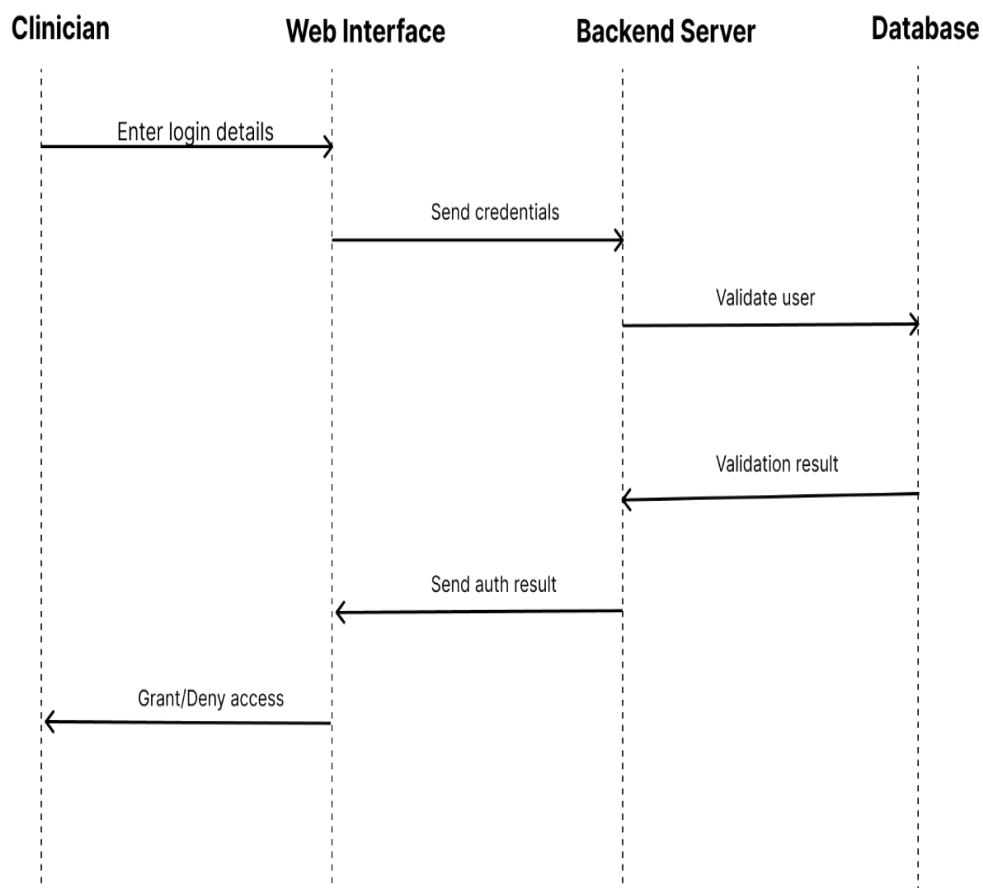


Figure 4.3: Authentication Sequence

These details are sent to the backend server, which verifies the credentials by checking stored records in the database. The database returns a success or failure response, which the backend forwards back to the user through the web interface. If the credentials are valid, access is granted and the user is directed to the dashboard; otherwise, the system denies access and prompts the

user to retry. This ensures that only authorized users can access sensitive medical data and system features.

Phase 2: Scan Upload and Diagnosis

This diagram represents the process of uploading a fundus scan and generating a diagnosis. The user begins by uploading a fundus image through the web interface.

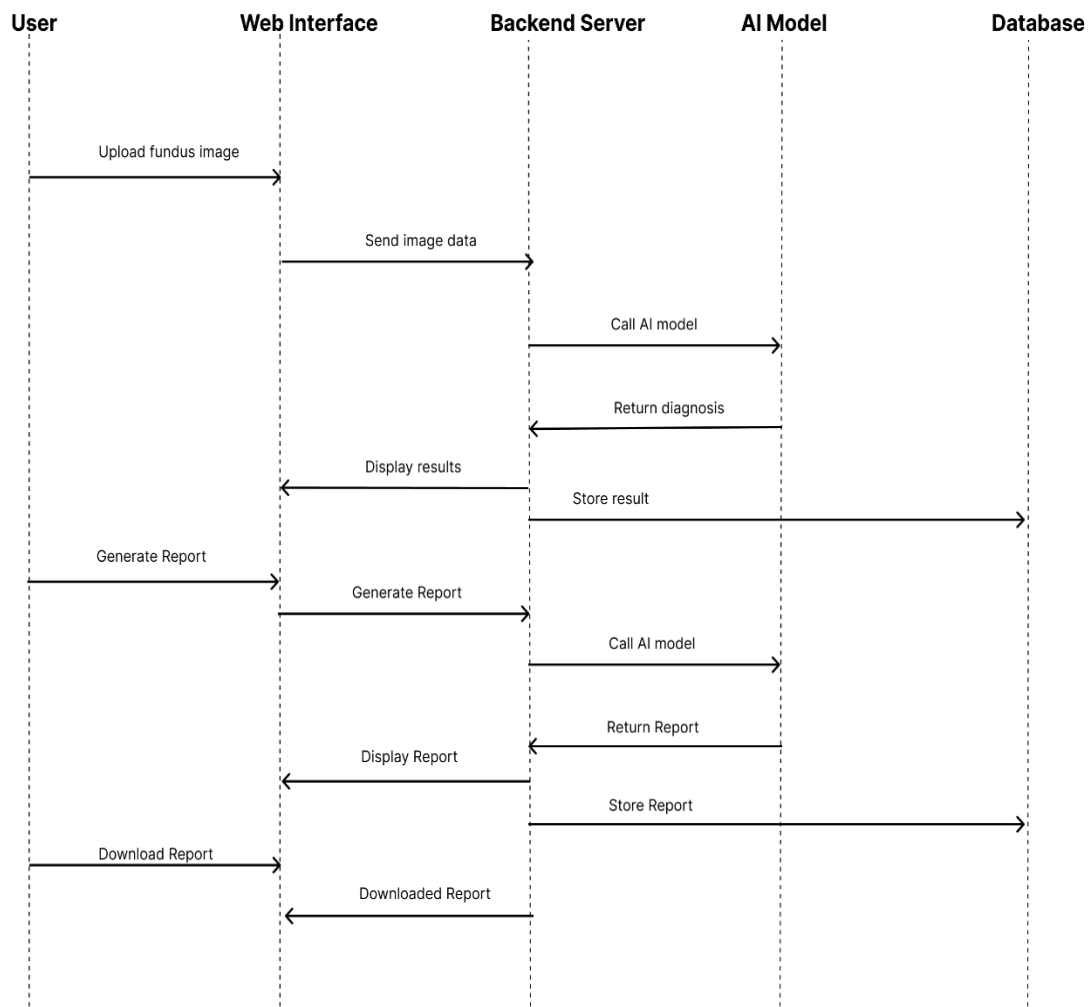


Figure 4.4: Scan Upload and Diagnosis Sequence Diagram

The image data is forwarded to the backend server, which sends it to the AI model for glaucoma detection. The model processes the scan and returns a prediction result. The backend then stores the diagnosis along with the scan data in the database. Once stored, the backend sends the final result and report back to the web interface, where it is presented to the user. This phase demonstrates the automated pipeline from scan upload to AI-powered diagnosis and report generation.

Phase 3: Dashboard, Reports, and Additional Features

This sequence diagram illustrates how users interact with the dashboard after authentication.

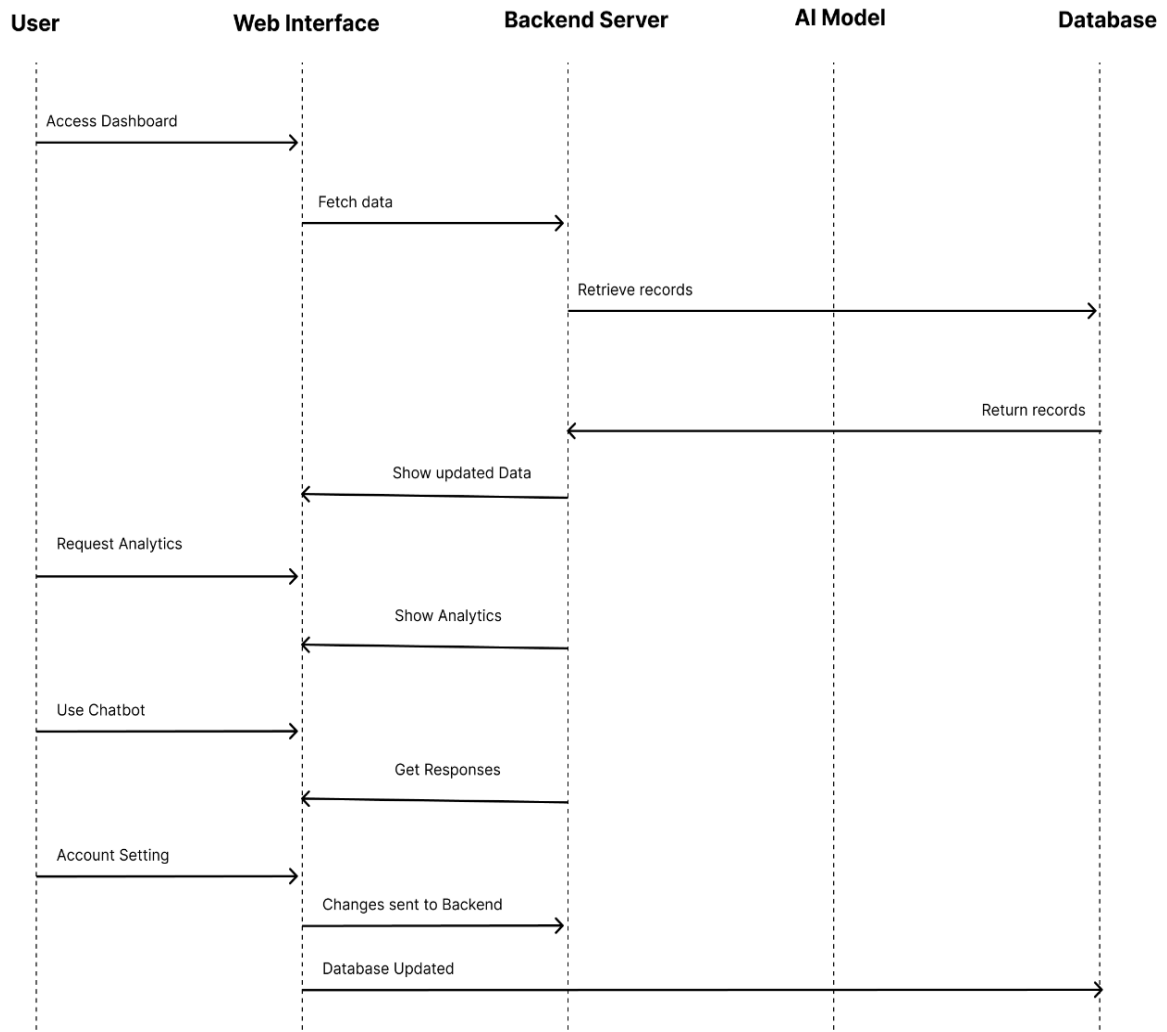


Figure 4.5: Dashboard and Reports Sequence Diagram

When a user requests information such as Patient History, Diagnostic Reports, Recent Scans, Analytics, or Account Settings, the web interface sends the request to the backend server. The backend retrieves the relevant data from the database and returns it to the web interface, where it is displayed to the user. For chatbot-related queries, the system forwards the request to the backend, which processes or relays the query to an external service before sending back the response. This phase highlights how the dashboard serves as a central hub for accessing all essential features of the system.

4.5 Class Diagram

The class diagram provides a structural representation of the Glaucoma Detection Web Application, illustrating its key classes, their attributes, operations, and the relationships between them. This diagram presents a clear overview of how different components work together to support scan upload, diagnosis processing, explanation generation, and report management within the system.

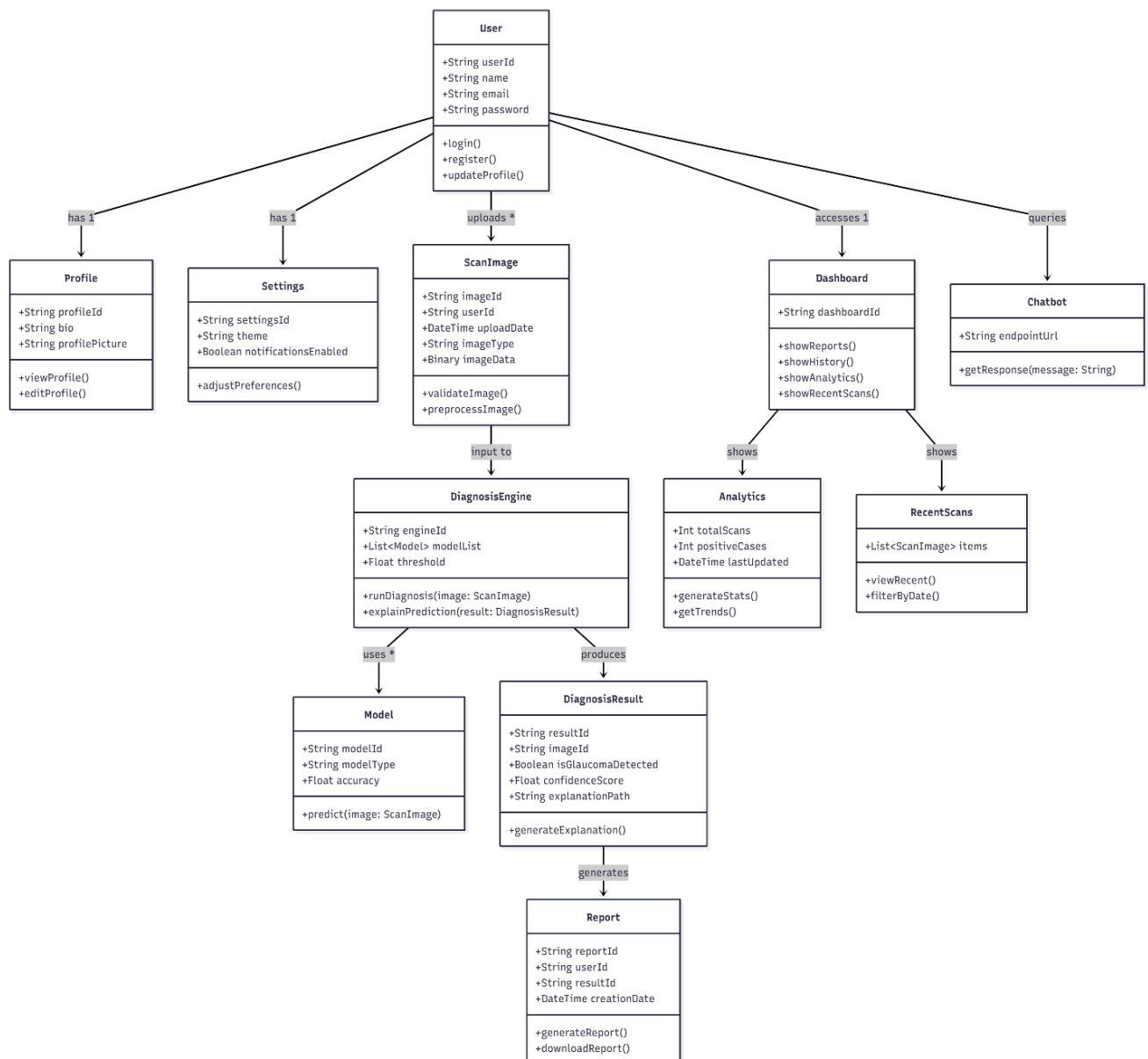


Figure 4.6: Glauco Vision Class Diagram

4.5.1 Components:

4.5.1.1. User

- Represents a clinician or patient using the system.
- Contains identifying information such as userId, name, email, and password.
- Provides core functionalities such as logging in, registering, and updating personal profile details.

4.5.1.2. Profile

- Stores extended user information including biography and profile picture.
- Allows viewing and editing profile information.

4.5.1.3. Settings

- Represents user-specific preferences such as theme selection and notification settings.
- Enables users to adjust system behavior according to their needs.

4.5.1.4. ScanImage

- Represents an uploaded fundus scan used for glaucoma analysis.
- Stores metadata such as imageId, uploadDate, scan type, and raw image data.
- Includes functions for validating and preprocessing images before diagnosis.

4.5.1.5. DiagnosisEngine

- Core diagnostic component of the system.
- Uses multiple ML models to analyze uploaded fundus images.
- Provides functions for running the diagnosis and generating visual explanations.

4.5.1.6. Model

- Represents an individual AI model used within the diagnosis engine.
- Includes model attributes and a method for predicting diagnostic output from a scan.

4.5.1.7. DiagnosisResult

- Stores the outcome produced by the diagnosis engine.
- Contains information such as glaucoma detection status, confidence score, and explanation file path.
- Supports generating explanation data for interpretability.

4.5.1.8. Report

- Represents the final diagnostic report generated for the user.
- Contains references to the associated user and diagnosis result.
- Allows generating and downloading complete diagnostic reports.

4.5.1.9. Dashboard

- Serves as the main access point after login.
- Provides options to view reports, patient history, analytics, and recent scans.

4.5.1.10. Analytics

- Provides statistical insights such as total scans and number of positive glaucoma detections.
- Includes methods for generating system trends and performance metrics.

4.5.1.11. RecentScans

- Displays a list of recently uploaded fundus images.
- Allows users to preview or filter recent scans by date.

4.5.1.12. Chatbot

- A lightweight chatbot API used to assist users by responding to queries.
- Generates basic automated responses for guidance and system usage support.

4.5.2 Relationships:

The following relationships describe how classes interact within the system:

- **User → Profile:** Each user has exactly one profile containing extended personal information.
- **User → Settings:** Every user is associated with one settings object to configure preferences.
- **User → ScanImage:** A user may upload multiple scans, forming a one-to-many relationship.
- **ScanImage → DiagnosisEngine:** Scan images are passed as input to the diagnosis engine.
- **DiagnosisEngine → Model:** The engine uses one or more AI models to analyze images.
- **DiagnosisEngine → DiagnosisResult:** The engine produces a diagnosis result object after processing.
- **DiagnosisResult → Report:** Each diagnosis result is used to generate a final downloadable report.
- **User → Dashboard:** Users interact with the dashboard after successful login.
- **Dashboard → Analytics:** The dashboard retrieves analytical summaries for display.
- **Dashboard → RecentScans:** The dashboard provides access to recently uploaded scans.
- **User → Chatbot:** Users may query the integrated chatbot for support or guidance.
- **Society and Follower:** Societies have followers, which represent the users who are part of or interested in the society.

4.5.3 Summary:

This class diagram outlines the structural foundation of the glaucoma detection system, illustrating how users interact with the application across scanning, diagnosis, reporting, and auxiliary features such as analytics and chatbot support. The relationships among classes ensure smooth data flow and a cohesive user experience, ranging from image upload to AI-generated diagnosis and report retrieval. By modularizing the system through clearly defined classes, the architecture promotes scalability, maintainability, and ease of future enhancements.

Chapter 5

Implementation

5 Implementation

In the implementation phase of the CUI Sphere app, we developed key algorithms that ensure smooth and effective functioning of the app, allowing university students, society heads, and administrators to interact seamlessly.

5.1 Algorithm

5.1.1.1 User Authentication

Function: authenticate User (username, password)

Input: username (string), password (string)

Output: user (object) or error (string)

Pseudocode:

1. Retrieve user record from the database based on the provided username.
2. If a user with the given username exists:
 - a. Compare the provided password with the stored password hash.
 - b. If the passwords match, return the user object.
 - c. If the passwords do not match, return an error message indicating "Incorrect password".
3. If no user is found, return an error message indicating "User not found".

5.1.1.2 Upload Fundus Image

Function: uploadFundusImage(user, imageFile)

Input: user (object), imageFile (file: JPG/PNG)

Output: scanRecord (object) or error (string)

Pseudocode:

1. Verify that user is authenticated.
2. If authorized:
 - a. Check that imageFile format is JPG or PNG.
 - b. Check that file size is within the allowed limit (e.g., ≤ 20 MB).
 - c. If validation passes, store the image in secure storage.

- d. Create a new scan record in the database with status "Pending Analysis".
- e. Return the scanRecord object.
3. If validation fails at any step, return an appropriate error message (e.g., "Unsupported format" or "File too large").
4. If user is not authenticated, return error "User not authenticated".

5.1.1.3 Run Glaucoma Diagnosis

Function: runDiagnosis(scanRecord)

Input: scanRecord (object containing image path and metadata)

Output: diagnosisResult (object) or error (string)

Pseudocode:

1. Load the fundus image from storage using the path in scanRecord.
2. Preprocess the image (resize, normalize, crop optic disc region if required).
3. Send the preprocessed image to the AI model inference service.
4. Receive model output:
 - a. predictionLabel (e.g., "Glaucoma" or "Healthy Eye").
 - b. confidenceScore (float between 0 and 1).
 - c. Return success.
5. Update the corresponding scan record in the database with the diagnosis and confidence score, set status to "Completed".
6. Create a diagnosisResult object containing label, confidence, and timestamp.
7. Return diagnosisResult.
8. If the model service fails or no response is received, return error "Diagnosis temporarily unavailable".

5.1.1.4 Generate Explainability Map

Function: generateExplainability(scanRecord)

Input: scanRecord (object with image path and diagnosis info)

Output: heatmapPath (string) or error (string)

Pseudocode:

1. Ensure that a completed diagnosis exists for scanRecord.
2. Load the original fundus image associated with the scan.
3. Pass the image and the model to an explainability method (e.g., Grad-CAM).
4. Receive a heatmap that highlights important regions used by the model.
5. Overlay the heatmap on the original image to create a visual explanation.
6. Save the resulting explainability image and store its path in the database.
7. Return heatmapPath.
8. If no diagnosis exists or generation fails, return an appropriate error message.

5.1.1.5 Generate Diagnostic Report

Function: generateReport(user, scanRecord)

Input: user (object), scanRecord (object including diagnosis and explainability path)

Output: reportFile (file path) or error (string)

Pseudocode:

1. Verify that user is authorized to view this scan (owner patient or assigned clinician).
2. If authorized:
 - a. Retrieve patient details and clinician information if available.
 - b. Retrieve diagnosis label, confidence score, and explainability image.
 - c. Create a structured report template (PDF/HTML).
 - d. Insert patient data, scan date, diagnosis summary, confidence, and clinical notes section.
 - e. Embed the explainability image into the report.
 - f. Save the report and store its path in the database.
 - g. Return reportFile path.
3. If user is not authorized, return error "Access denied for this report".

5.1.1.6 Retrieve Recent Scans

Function: getRecentScans(user, limit)

Input: user (object), limit (integer – number of scans to fetch)

Output: scanList (array of objects) or error (string)

Pseudocode:

1. Confirm that user is authenticated.
2. Query the database for scans belonging to the user, ordered by date (most recent first).
3. Limit the results to the requested limit.
4. For each scan, select key fields (scan ID, date, diagnosis label, confidence).
5. Return scanList.
6. If no scans exist, return an empty list or message "No scans found".

5.1.1.7 Chatbot Interaction

Function: askChatbot(user, query)

Input: user (object), query (string)

Output: response (string)

Pseudocode:

1. Check if user is logged in.
2. If logged in::
 - a. Log the query with a timestamp for future reference.
 - b. Send the query to the chatbot service with context.
 - c. Receive the chatbot response text.
 - d. Append a disclaimer if the response is health-related (e.g., "Not a substitute for professional medical advice").
 - e. Return the formatted response.
3. If not logged in, return error "User not authenticated".

These are simplified pseudocode representations of the algorithms for each component and function of our GlaucoVision website. The actual implementation includes additional input validation, error handling, security checks, and database interactions to ensure the system runs reliably and safely in a real clinical environment. Furthermore, backend integration and model-response handling have been optimized to support real-time AI-assisted screening.

Table 5.13: External APIs

Name of API	Description of APIs	Purpose of Usage	Function/Class Name
OpenRouter Chat Completions API (Mistral-7B)	Provides cloud-based large language model responses over HTTPS. Allows sending a medical query as text and receiving a natural-language reply.	Powers the GlaucoVision medical chatbot to answer health-related questions for patients and clinicians in an educational, non-diagnostic way.	Chatbot component, sendMessage, stripReasoning, typewriterAppend

5.2 User interface

The user interface (UI) of GlaucoVision is designed to provide a smooth, medical-grade experience tailored for both patients and clinicians. The interface follows a clean, structured, and modern design language, ensuring accessibility, clarity, and easy navigation. Below are the main screens:

- **Figure 5.1:** The Sign In screen where users securely enter their email and password to access personalized dashboards, scan reports, and analysis history.
- **Figure 5.2:** The Sign Up screen allowing new users to create an account by entering basic details and selecting their role as Patient or Clinician.
- **Figure 5.3:** The Home Page displaying the platform’s introductory content, key features, workflow explanation, testimonials, pricing plans, and common FAQs.
- **Figure 5.4:** The Upload Scan screen where users can drag-and-drop retinal fundus images or browse files for AI-based glaucoma detection.
- **Figure 5.5:** The Features Overview section highlighting accuracy, detailed reports, advanced analytics, and security measures.
- **Figure 5.6:** The Step-by-Step Workflow module explaining the complete process from image upload to diagnostic report generation.
- **Figure 5.7:** The Testimonials screen presenting real user feedback from ophthalmologists, optometrists, and patients.

- **Figure 5.8:** The Pricing Plans screen showing Free and Premium tiers with their respective capabilities.
- **Figure 5.9:** The FAQ screen addressing important queries regarding diagnostic accuracy, data handling, supported formats, and limitations.
- **Figure 5.10:** The Dashboard showing total scans, recent results, average confidence, announcements, and quick action options.
- **Figure 5.11:** The Analytics screen providing detailed visual insights through charts and graphs.
- **Figure 5.12:** The Recent Scans screen listing all scan histories with timestamps and AI-detected status.
- **Figure 5.13:** The Chatbot interface enabling users to receive general health advice and medical support.
- **Figure 5.14:** The Account Settings screen where users update profile information, avatar, email, and account type.
- **Figure 5.15:** The Billing screen displaying subscription details, current plan, and upgrade options.
- **Figure 5.16:** The Logout screen confirming the user's intent to sign out securely.

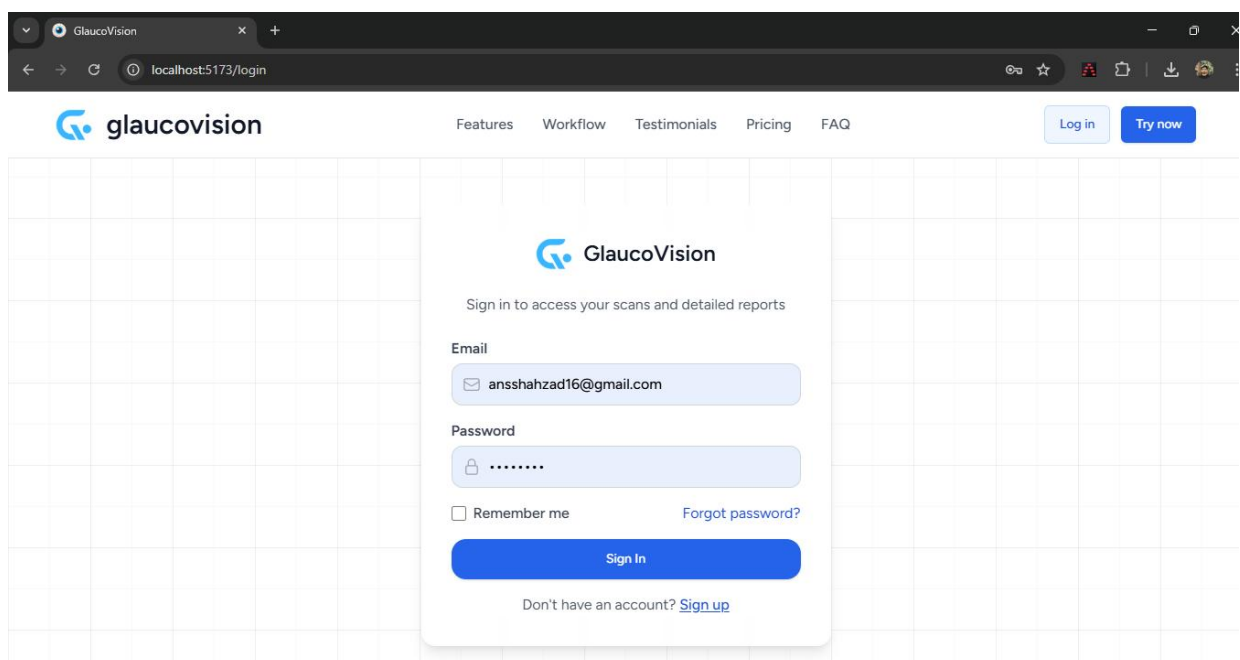


Figure 5.1: GlaucoVision Sign In Screen

The Sign In screen serves as the secure entry point where users authenticate themselves before accessing the GlaucoVision platform. At the top, the GlaucoVision logo is prominently positioned, reinforcing brand identity and establishing trust. The central panel displays two neatly aligned text fields, one for email and the other for password, each accompanied by subtle placeholder text and icons to enhance clarity and guide user input. A “Forgot Password?” link provides an immediate pathway for users who may have misplaced their credentials. The “Sign In” button is highlighted in a bold, professional blue color, drawing user attention as the primary action. Beneath it, a prompt for new users directs them to create an account, ensuring seamless navigation between authentication pages. The overall layout is clean, distraction-free, and optimized for both desktop and mobile use, ensuring a smooth, secure, and user-friendly login experience.

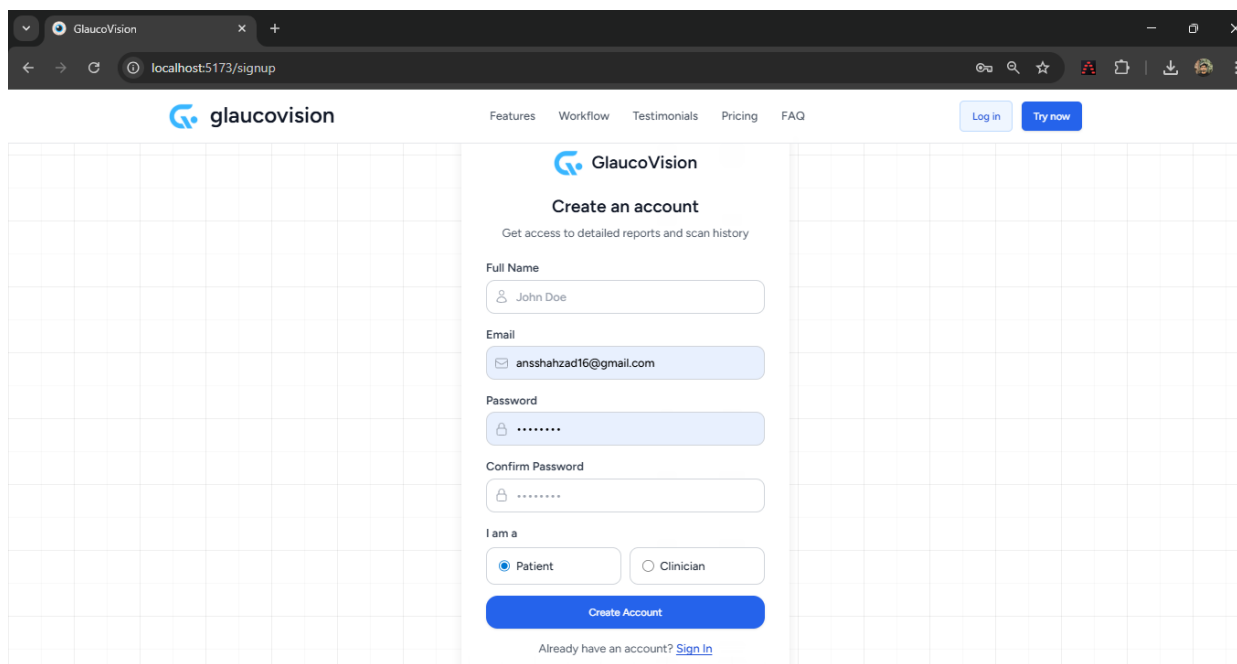


Figure 5.2: GlaucoVision Sign Up Screen

The Sign Up screen allows new users to register and begin using the platform. A clean header featuring the GlaucoVision branding gives the screen a professional and welcoming appearance. The layout contains structured form fields, including Name, Email, Password, and Account Type, all spaced evenly to avoid visual clutter. Icons and placeholder guidance help reduce input errors and enhance user interaction. One of the most important components is the role selection dropdown, allowing users to choose between “Patient” and “Clinician,” which tailors the dashboard and analytics features to their profile. The “Create Account” button is large and visually

appealing, emphasizing the key action. At the bottom, a link allows existing users to transition back to the Sign In screen effortlessly. This page ensures easy onboarding while maintaining clarity and reducing cognitive load for new users.

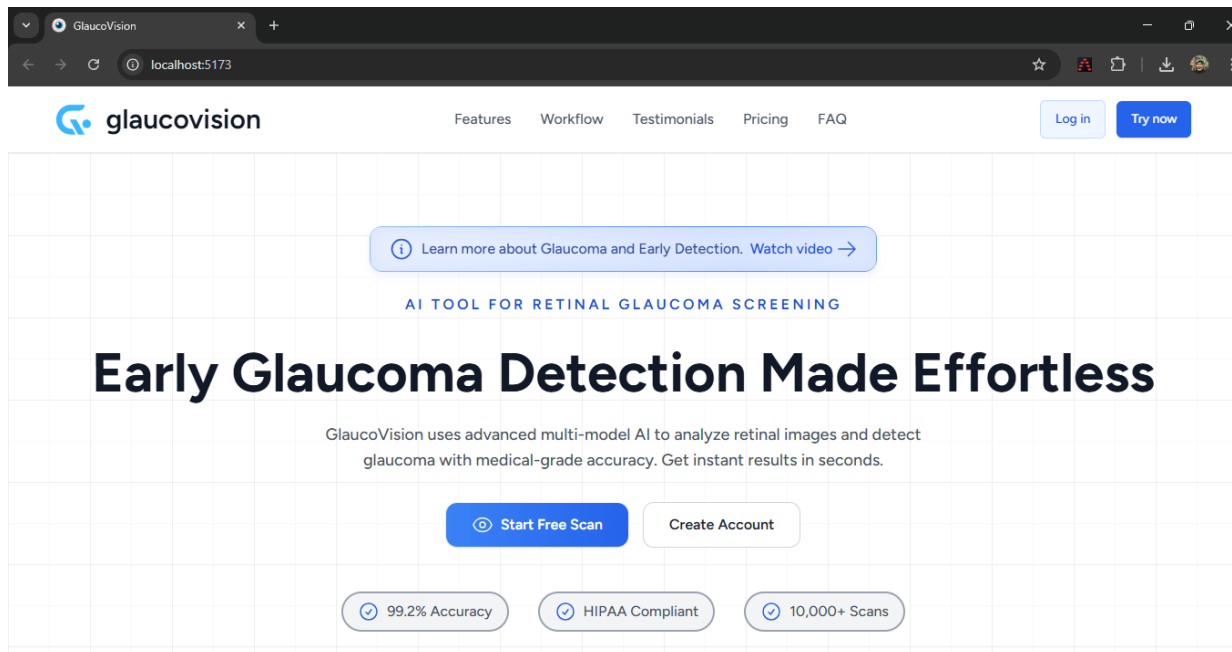


Figure 5.3: GlaucoVision Home Screen

The Home Page is the primary landing screen after login and introduces users to the features and capabilities of GlaucoVision. The top navigation bar includes links to Features, Workflow, Testimonials, Pricing, and FAQs, allowing users to quickly explore essential system information. The hero banner emphasizes the platform’s purpose—AI-powered glaucoma screening - with a concise yet impactful tagline and a large “Try Now” or “Start Free Scan” button to initiate the scanning process. As users scroll, they encounter well-organized sections highlighting key strengths such as accuracy, speed, and accessibility. The workflow section visually breaks down how GlaucoVision processes retinal images, making the underlying AI operations easier to understand. Testimonials from clinicians and patients build trust, while pricing cards clearly differentiate the Free and Premium plans. Overall, the Home Page combines educational and promotional elements to guide users smoothly through system exploration.

Overall, the Home Page combines educational content, visual clarity, promotional elements, and intuitive navigation to ensure a smooth introduction to GlaucoVision. Whether the user is a first-time visitor, a patient concerned about eye health, or a clinician seeking advanced diagnostic support, the Home Page effectively communicates value.

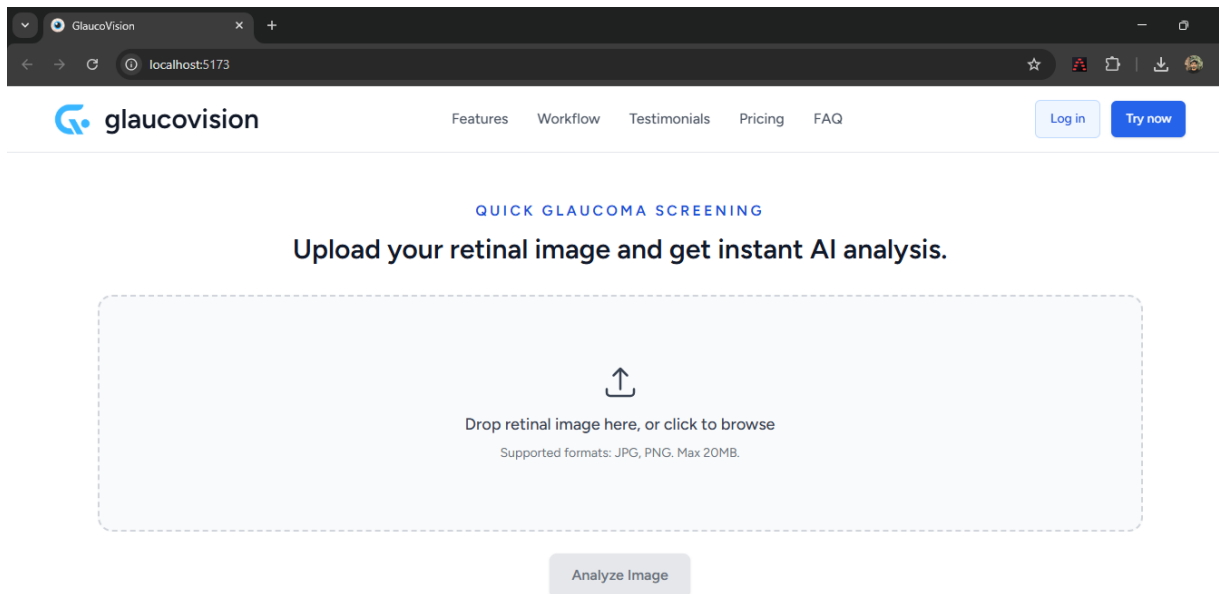


Figure 5.4: GlaucoVision Upload Screen

The Upload Scan screen is central to GlaucoVision’s diagnostic workflow, enabling users to upload retinal fundus images for AI-based glaucoma detection. At the center of the screen, a large drag-and-drop upload box is outlined with a dotted border, visually guiding users to place their retinal scan images. Alternatively, a “Browse Files” option is available for users unfamiliar with drag-and-drop interactions. Once an image is selected, a preview may appear to confirm the correct file has been chosen. A prominently styled “Analyze Scan” button activates after a valid image is uploaded, encouraging users to initiate AI processing.

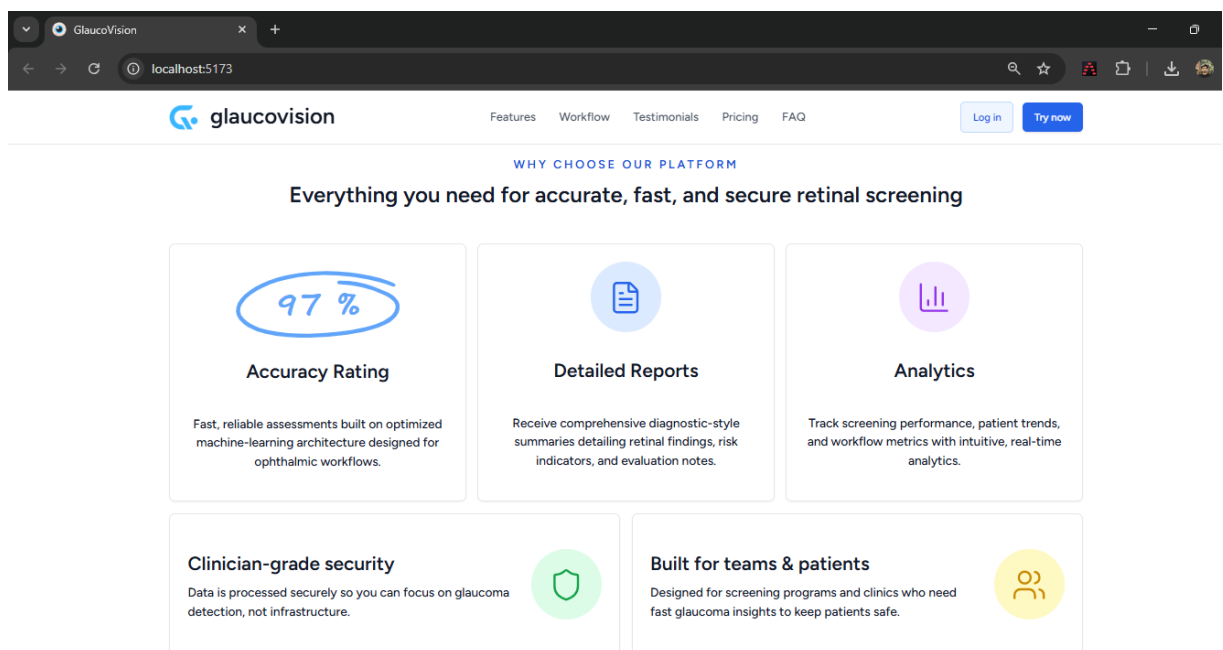


Figure 5.5: GlaucoVision Features Screen

This screen highlights GlaucoVision’s core features through clean, card-style sections, each incorporating an icon and short descriptive text. These features include high diagnostic accuracy, detailed AI-generated reports, secure data storage, and trend-based analytics for long-term monitoring. The layout is crafted to emphasize professionalism and medical reliability, using consistent color accents and spaced-out elements that improve readability. Each feature card visually stands out, helping users understand the platform’s technical benefits at a glance. The combination of icons and concise explanations enhances user comprehension and communicates credibility. This screen plays a key role in educating new users about how GlaucoVision enhances glaucoma screening compared to traditional methods.

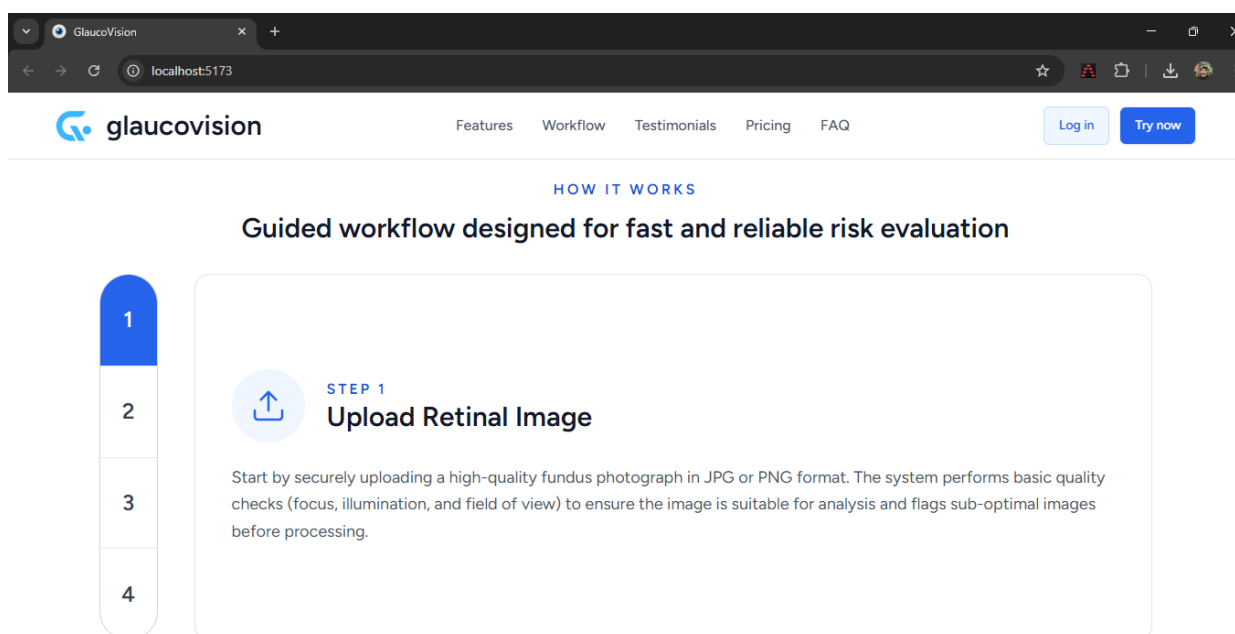


Figure 5.6: GlaucoVision Workflow Screen

The Workflow screen provides a step-by-step visual demonstration of how GlaucoVision processes a retinal scan to produce diagnostic results. A vertical or horizontal step indicator is used to show each stage of the pipeline, beginning with Image Upload, followed by AI Processing, Result Generation, and finally Report Download. Each step includes descriptive text explaining what happens in the backend - for example, how the AI evaluates optic nerve features and computes the cup-to-disc ratio. Icons and smooth transitions make the explanation visually engaging and easier to follow. This screen helps users especially first-time visitors understand the technical workflow without needing medical or AI-related background knowledge.

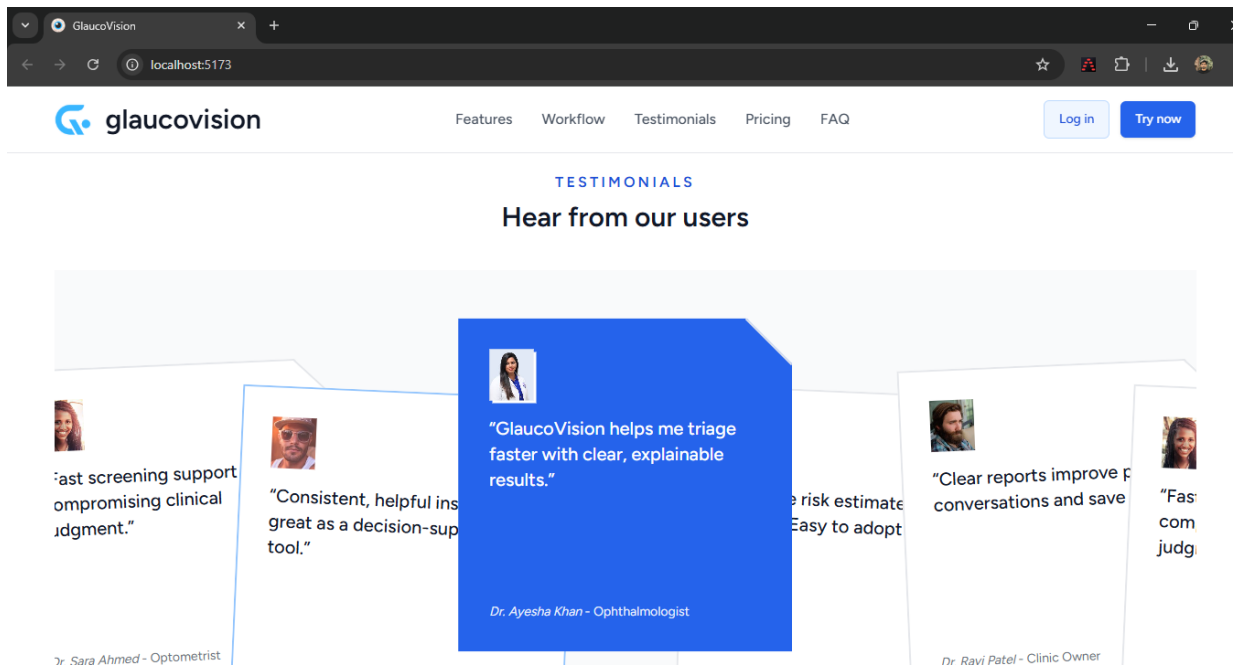


Figure 5.7: GlaucoVision Testimonials Screen

The Testimonials section showcases feedback from clinicians, ophthalmologists, optometrists, and patients who have used GlaucoVision. Each testimonial card includes a profile picture, user name, and role, giving authenticity and personalization to the feedback. The main body of the testimonial provides insight into user satisfaction regarding accuracy, usability, and trust in the system's AI predictions. A slider or grid layout ensures visual structure while allowing multiple testimonials to be displayed without overwhelming the user.

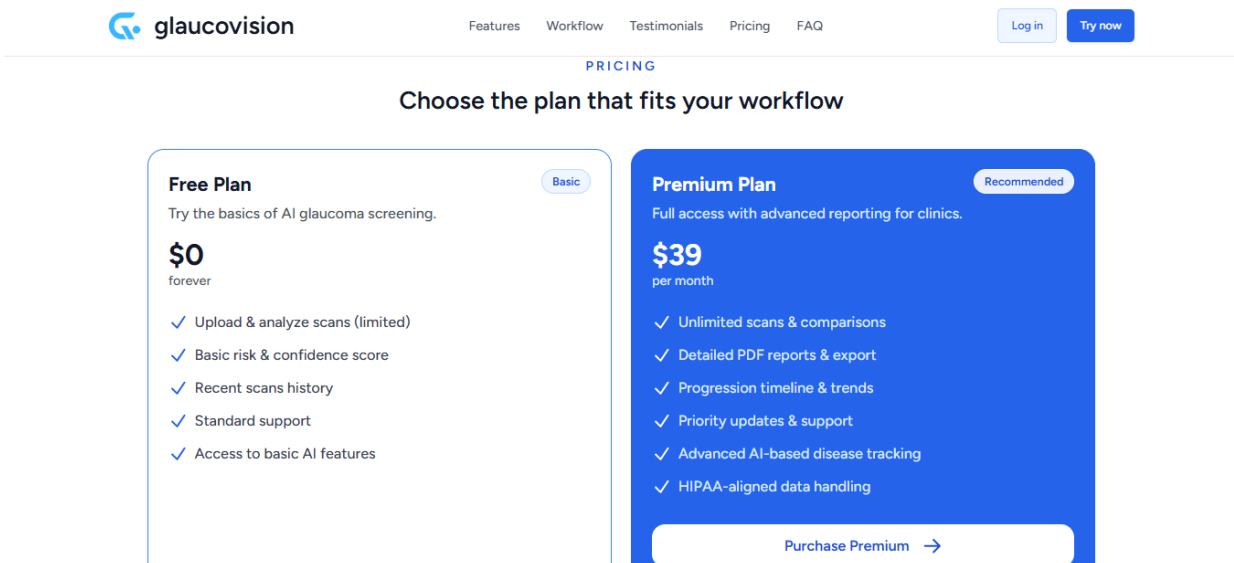


Figure 5.8: GlaucoVision Pricing Screen

This screen presents the available subscription plans using a side-by-side comparison layout, making it easy for users to evaluate their options. The Free Plan highlights essential features such as limited scans, basic reports, and standard support, while the Premium Plan emphasizes advanced capabilities like unlimited scans, PDF exports, extended AI analytics, and priority support. The Premium card is visually emphasized through a bolder color scheme and a “Recommended” badge to guide user decisions. Each plan includes a bullet-point list of benefits, pricing information, and a clearly labeled button such as “Upgrade to Premium.” The screen provides transparency and helps users select the plan that aligns with their needs

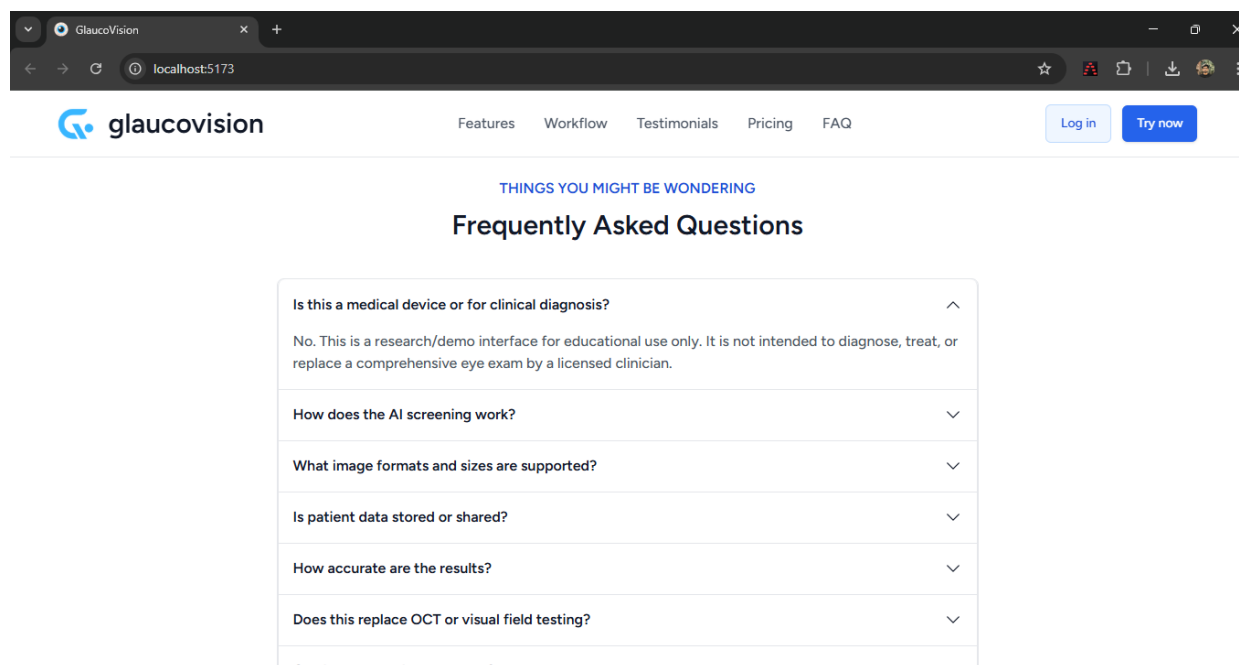


Figure 5.9: GlaucoVision FAQ Screen

The FAQ screen addresses important questions users may have about glaucoma detection accuracy, supported image formats, privacy policies, AI limitations, and general usage. The page uses an accordion-style interface where each question expands upon clicking, revealing detailed explanations underneath. This approach maintains a clean layout while providing comprehensive information. The tone remains professional and reassuring, especially when explaining clinical limitations and emphasizing that GlaucoVision assists rather than replaces medical professionals. This section reduces user confusion and builds trust by offering transparent, easy-to-understand guidance. The expandable design also improves accessibility and usability across different devices. Clear explanations about data handling and security further reinforce user confidence. Overall, the FAQ section enhances user satisfaction by providing reliable support and setting realistic expectations about the system’s capabilities.

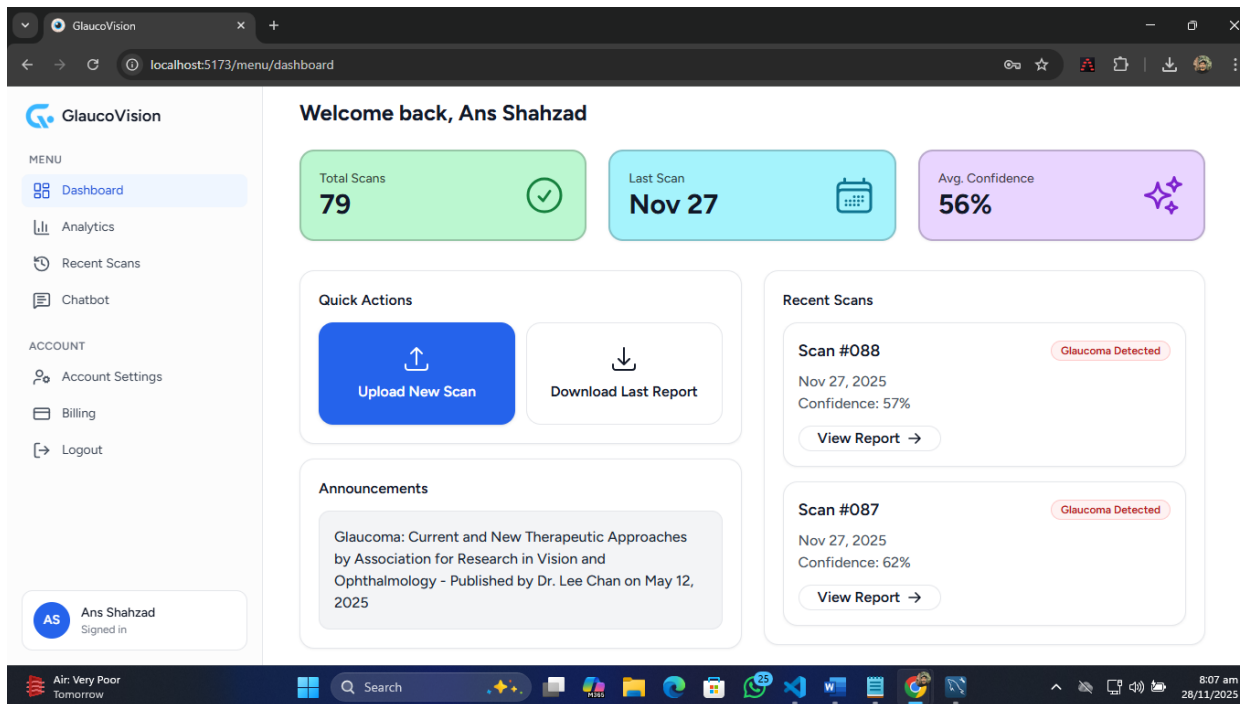


Figure 5.10: GlaucoVision Dashboard Screen

The Dashboard is the central workspace for both patients and clinicians, offering a comprehensive overview of user activity, system insights, and recently analyzed scans. Its layout is designed to present critical information at a glance while maintaining a clean, structured, and visually engaging interface. At the top of the screen, users are greeted with a row of metric cards that summarize key performance indicators such as Total Scans Performed, Last Scan Date, AI Confidence Levels, and Risk Tracing Progress. Each card is color-coded and icon-supported, enabling users to quickly differentiate between categories and interpret key values without confusion. These cards not only provide a quick snapshot of usage history but also help users monitor long-term trends in their eye health or diagnostic workload.

The entire dashboard emphasizes clarity, accessibility, and usability. Thoughtful spacing, modern typography, and carefully selected accent colors contribute to a polished visual experience while ensuring that the content remains easy to interpret. Whether viewed on a desktop or tablet, the Dashboard provides immediate insight into analytical trends, supports efficient navigation across the platform, and enhances the overall user journey. In summary, the dashboard acts as a central control hub, empowering users with actionable information, seamless workflow shortcuts, and continuous system feedback—all essential elements in an AI-assisted glaucoma screening environment.

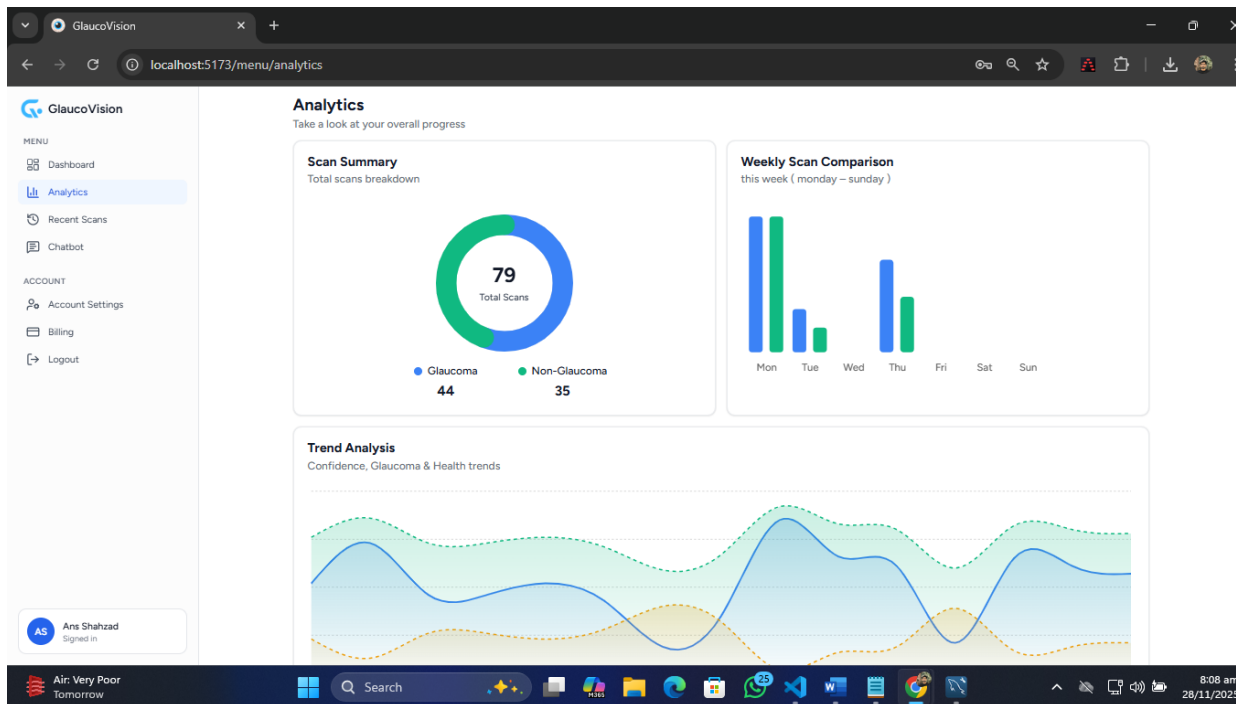


Figure 5.11: GlaucoVision Analytics Screen

The Analytics screen serves as one of the most informative and insight-driven components of the GlaucoVision platform. Designed with a clean and spacious visual layout, this screen translates raw diagnostic data into meaningful patterns and trends that are easy to interpret for both patients and clinicians. At the top, a prominent circular chart summarizes the overall distribution of glaucoma-positive versus non-glaucoma scans. This chart uses distinct colors and precise percentages, enabling users to instantly understand the proportion of risk cases within their scan history. For clinicians, this helps assess screening effectiveness across different patient groups, while patients can quickly visualize their personal diagnostic trends.

The Analytics screen carefully balances visual richness with interpretability. Each graph is spaced thoughtfully, ensuring that the interface does not feel crowded despite displaying multiple layers of information. Hover interactions and tooltips (if implemented) provide additional details when needed, further enhancing the user experience without overwhelming beginners.

For clinicians, this screen functions as a valuable diagnostic support tool helping them detect trends that might not be immediately evident from individual scan reports. It aids in identifying early-stage risk patterns, monitoring treatment influence, and validating screening consistency. For patients, the visualizations offer reassurance and clarity by transforming complex medical predictions into easy-to-understand graphical insights.

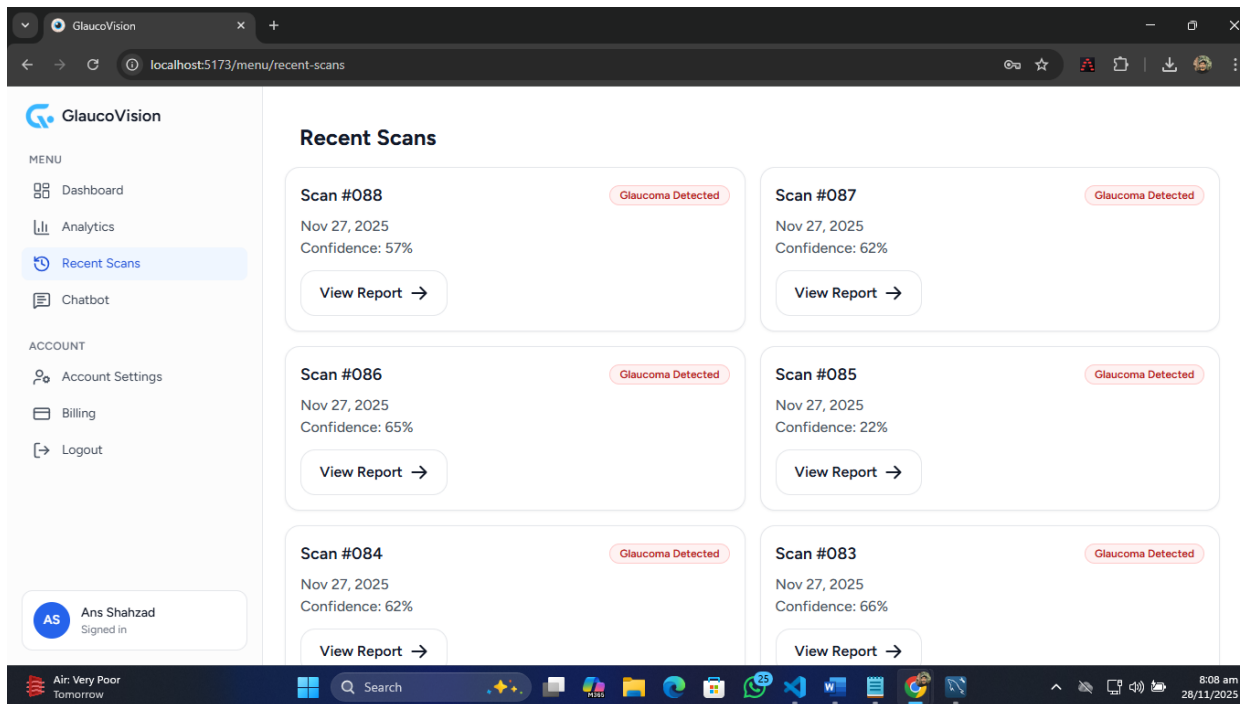


Figure 5.12: GlaucoVision Recent Scans Screen

The Recent Scans screen compiles all previous scans into a neatly organized list. Each scan appears as an individual card displaying scan ID, date, confidence score, and AI-generated disease status with a clear label such as “Glaucoma Detected” or “Healthy.” A “View Report” button allows users to revisit detailed results and download their diagnostic summary. This screen ensures quick access to historical records, which is especially important for clinicians monitoring progression or patients tracking their eye health over time. The consistent layout and spacing make browsing intuitive and efficient.

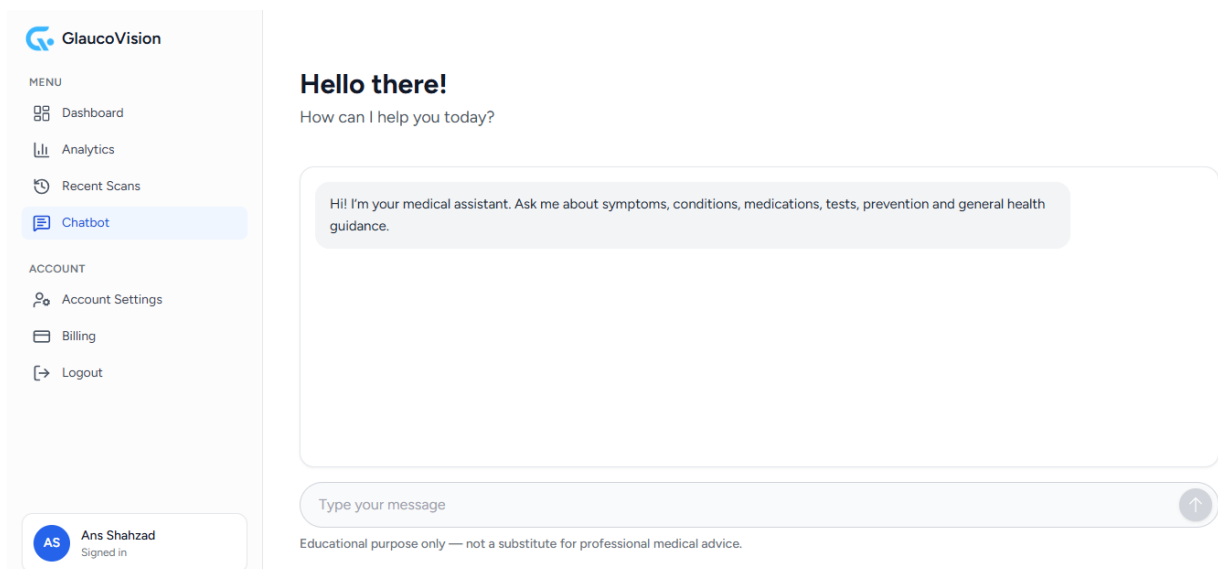


Figure 5.13: GlaucoVision Chatbot Screen

The Chatbot interface provides users with general medical guidance related to symptoms, preventive care, medications, and eye health. The screen includes a welcoming message at the top, inviting users to interact. The main conversation area displays messages in distinguishable chat bubbles—grey for bot responses and blue for user inputs. At the bottom, a text input field allows users to type queries, with an arrow button to send messages. The design is minimal, clean, and friendly, ensuring users feel comfortable asking health-related questions. Although not a replacement for professional diagnosis, the chatbot provides helpful educational insights.

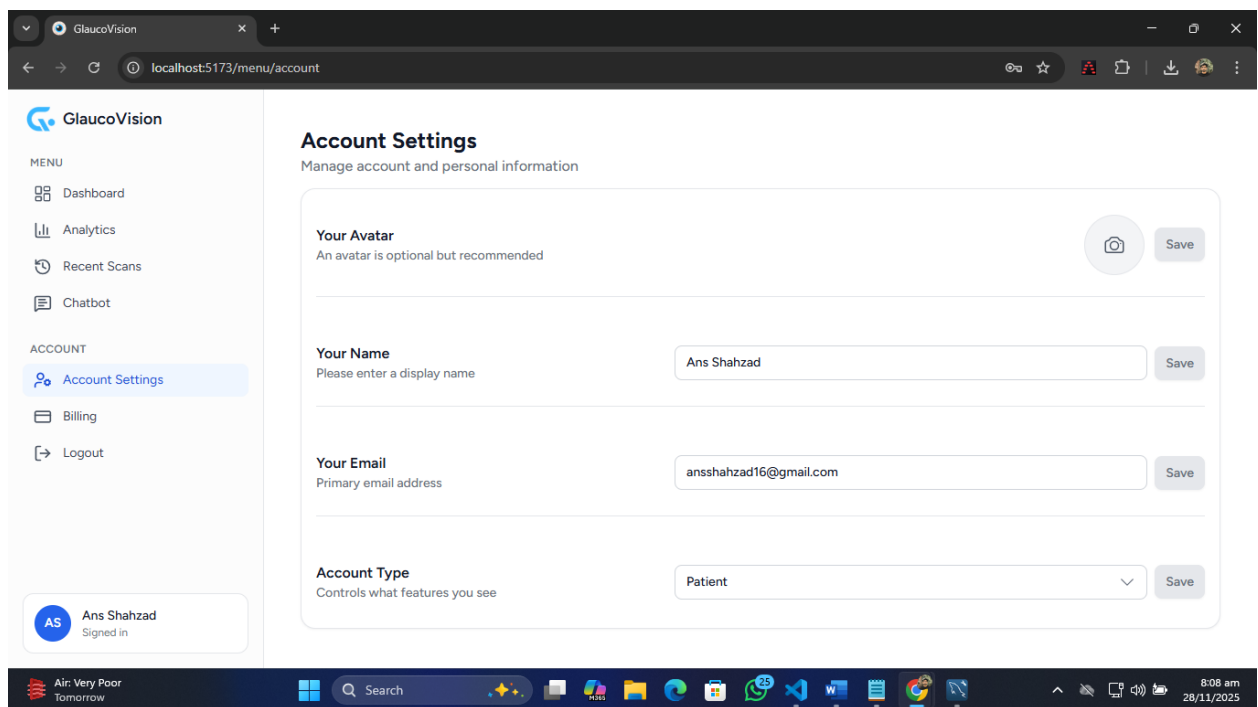


Figure 5.14: GlaucoVision Account Settings Screen

The Account Settings screen enables users to update personal information in a structured and secure environment. Separate sections allow modification of the profile avatar, name, email, and account type. Each section includes a dedicated “Save” button, reducing errors and enabling partial updates without affecting unrelated fields. A role dropdown allows switching between Patient and Clinician roles, depending on the user's permissions. The interface maintains wide spacing and clean typography, making data management simple and user-friendly. This screen plays an important role in maintaining accurate user information and customization preferences.

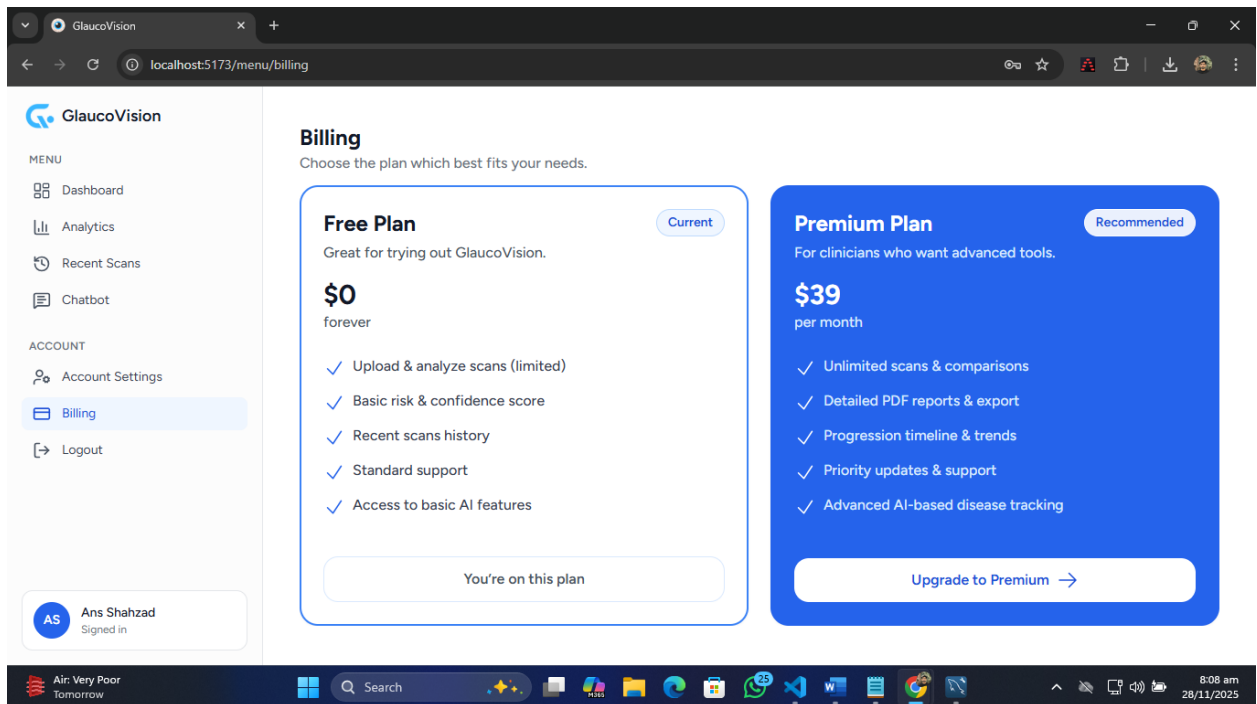


Figure 5.15: GlaucoVision Billing Screen

The Billing screen provides a clear, structured, and transparent overview of the user’s subscription status, ensuring that both patients and clinicians can easily understand their plan benefits and make informed upgrade decisions. At the top of the screen, the system displays the user’s current active plan, which is visually emphasized through a “Current” badge on the Free Plan card. This card summarizes the basic features available at no cost, such as limited scans, standard processing speed, and access to essential AI-generated results. The card uses soft colors and concise bullet points to make the included features easy to interpret at a glance.

A large “Upgrade to Premium” button appears below the plan details, serving as the primary call-to-action. When users click this button, they are securely redirected to the Stripe payment gateway, where all financial transactions are processed. Stripe ensures compliance with modern security standards, including encrypted card details, PCI-DSS compliance, and fraud protection mechanisms. On the payment page, users can enter their card information, review billing summaries, and confirm their subscription with a single secure checkout flow. After the transaction is completed, Stripe automatically redirects users back to GlaucoVision, where their dashboard updates to reflect the Premium Plan status.

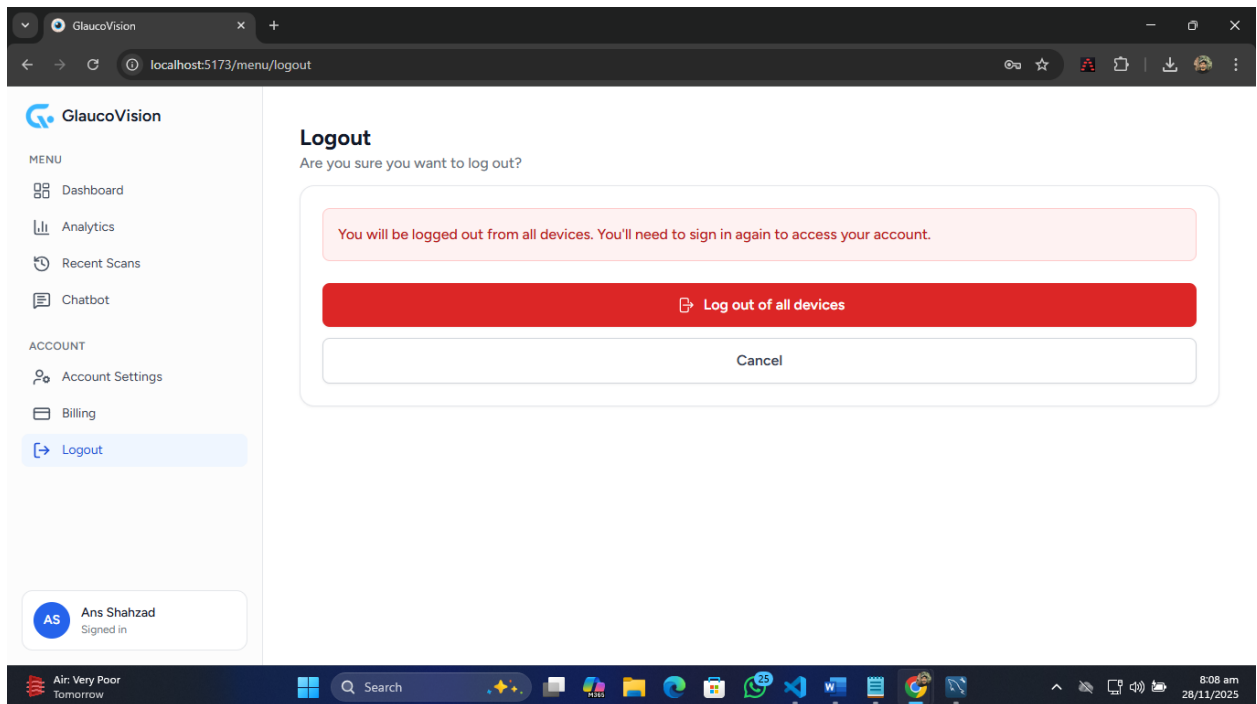


Figure 5.16: GlaucoVision Logout Screen

The Logout screen ensures secure session termination by confirming the user’s decision before logging them out. A red warning box informs the user that logging out will end the session across all connected devices, reinforcing security awareness. The primary red “Log out of all devices” button is bold and attention-grabbing, while a secondary “Cancel” button allows users to safely return if they triggered the logout accidentally. The interface is intentionally simple, removing distractions to ensure that users make a deliberate and informed choice.

Chapter 6

Testing and Evaluation

6 Testing and Evaluation

Testing ensures whether the required functionality is developed and meets user needs. It is performed at different phases, with errors identified and resolved before proceeding. Testing evaluates the program's ability to meet desired results. In many software engineering methodologies, it is a separate phase after implementation, allowing fresh perspectives to identify errors more effectively.

6.1 Manual Testing

Manual Testing is a type of Software Testing where testers manually execute test cases without automation tools. It is the most basic testing method, used to find software bugs. Testers act as end users, exploring the application's features to ensure correct behavior, though they may have limited insight into the product's inner workings, potentially missing deeper issues or root causes [9].

6.1.1 System Testing

System testing ensures the application meets its requirements and quality standards. It is performed after development to verify the system works as intended and uncovers hidden errors. Types of testing include unit, functional, and integration testing, all of which must be completed before deployment.

6.1.2 Unit Testing

Unit testing in the GlaucoVision system focuses on testing individual components such as login authentication, fundus image upload, AI diagnosis, explainability generation, and report download independently to identify logic or functional errors. For example, it includes validating user inputs like email and password, checking image formats and file sizes, and confirming that buttons and forms perform their expected actions.

The main advantage of unit testing is that these tests can be run repeatedly whenever any part of the system changes, helping developers quickly detect and fix bugs. Developers rely on unit testing to confirm that each module (like authentication, image upload, or diagnosis generation) behaves correctly before moving into integration or user-level testing phases.

Unit Testing 1: Login as Clinician/Patient

Testing Objective: To ensure the login form correctly authenticates valid users and rejects invalid attempts.

Table 6.1.2.14: Login as Clinician/Patient

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Login with valid credentials	Username: ansshahzad16@gmail.com Password: mian.ans	User logs in and is redirected to dashboard	Pass
UT-002	Login with invalid credentials	Username: wrongadmin@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass
UT-003	Login with empty fields	Username: Password:	Prompt: "Please enter email and password"	Pass
UT-004	Login with unregistered account	Email not found in database	Error message: "User does not exist"	Pass

Unit Testing 2: Fundus Image Upload

Testing Objective: To ensure that users can upload valid fundus images and the system handles invalid files properly.

Table 6.1.2.15: Fundus Image Upload Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Upload valid fundus image	File: fundus_01.png (valid size + supported format)	Image accepted and prepared for diagnosis	Pass

UT-002	Upload invalid file format	File: myfile.pdf	Error message: "Upload JPG/PNG fundus image only."	Pass
UT-003	Upload image exceeding size limit	File > 40MB	Error message indicating size limit exceeded	Pass
UT-004	Upload with no file selected	No file chosen	Prompt: "Please select an image to upload."	Pass

Unit Testing 3: Diagnosis, Explainability & Report Generation

Testing Objective: To ensure the AI diagnosis pipeline works correctly for valid uploads, including prediction, explainability map, and report creation.

Table 6.1.2.16: Diagnosis & Report Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Generate diagnosis from uploaded image	Valid preprocessed fundus image	System returns prediction and confidence score	Pass
UT-002	Generate explainability map	Diagnosis result	Heatmap successfully created and displayed	Pass
UT-003	Download diagnostic report	Report generated	Report downloads successfully in PDF/HTML format	Pass
UT-004	Attempt report download without diagnosis	No diagnosis performed yet	Message: "No report available."	Pass

6.1.3 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the modules is tested. This is to ensure that the system produced meets the specifications and requirements.

Functional Testing 1: Login as Clinician

Objective: To ensure that the correct diagnostic dashboard and clinician-specific features are displayed and functional upon login.

Table 6.1.3.17: Login as a Clinician

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	Clinician login with valid credentials	Username: alimurad1911@gmail.com Password: AliMurad@123	Clinician successfully logs into the system	Pass
FT-002	Login with invalid credentials	Username: wrongadmin@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass
FT-003	Upload a new fundus image	Image file: JPG/PNG fundus image	Image successfully uploaded and displayed for analysis	Pass
FT-004	Perform glaucoma diagnosis	Click “Analyze” after image upload	AI model generates prediction and confidence level	Pass
FT-005	Generate explainability heatmap	Click “Explainability”	Heatmap generated and displayed correctly	Pass
FT-006	Generate diagnostic report	Input: Patient name, age, uploaded image, diagnosis	Report created with diagnosis and heatmap details	Pass
FT-007	Download diagnostic report	Click “Download Report”	Report successfully downloaded	Pass

FT-008	Update clinician profile	New name, password, or profile image	Profile updated successfully	Pass
FT-009	Click the logout button and logout	Logout	Clinician exits system and returns to login	Pass

The functional testing table outlines a series of essential test cases designed to validate the primary diagnostic features available to clinicians in the GlaucoVision system. Test Case FT-001 verifies the login process using valid clinician credentials, ensuring that authorized users can access the system without errors. Test Case FT-002 evaluates how the system handles incorrect login attempts, confirming that an appropriate “Invalid credentials” message is displayed when wrong information is entered. Test Case FT-003 tests the ability to upload a valid fundus image, and the system successfully processes and displays it for analysis.

Test Case FT-004 focuses on the AI-powered diagnosis process, where clicking the “Analyze” button triggers the model to generate a glaucoma prediction along with a confidence score; the system performs this operation correctly. Test Case FT-005 verifies the explainability feature by checking whether the heatmap highlighting important retinal regions is generated and displayed properly. In Test Case FT-006, clinicians create a diagnostic report that combines the patient’s details, AI prediction, and the generated explainability map. The system handles this process smoothly and produces a complete report.

Test Case FT-007 confirms that the report download functionality works as expected, allowing clinicians to save reports for clinical documentation. In Test Case FT-008, clinician profile updates—including changes to name, password, or profile image—are tested and reflected accurately in the system. Finally, Test Case FT-009 validates the logout process, ensuring that clicking the logout button properly ends the session and redirects the user to the login screen. All test cases passed successfully, demonstrating that the clinician workflow is stable, accurate, and fully aligned with functional requirements.

Functional Testing 2: Login as a Patient

Objective: To ensure that all patient-level features work correctly and provide smooth access to reports, explainability maps, and system tools.

Table 6.1.3.18: Login as a Patient

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	Patient Login with valid credentials	Username: ansshahzad16@gmail.com Password: mianans123	Patient successfully logs into the system	Pass
FT-002	Patient login with invalid credentials	Username: wrongpresi@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass
FT-003	View diagnostic history	Navigate to “My Reports”	All previous diagnostic reports are displayed	Pass
FT-004	Download diagnostic report	Click “Download Report”	Report downloaded successfully	Pass
FT-005	Use chatbot for general assistance	Query example: “What does my diagnosis mean?”	Chatbot responds with non-diagnostic educational information	Pass
FT-006	Update profile	Change password, name, or profile image	Profile updated successfully	Pass
FT-007	Click the exit button and logout	Logout	Patient successfully logs out and returns to login.	Pass

The functional testing table documents a series of essential test cases designed to evaluate the core features available to patients within the GlaucoVision system. Test Case FT-001 confirms that a patient with valid login credentials can successfully access the platform and proceed to the dashboard. Test Case FT-002 verifies that incorrect login attempts are handled properly, with the system displaying the expected “Invalid credentials” message to prevent unauthorized access.

Test Case FT-003 evaluates access to the patient’s diagnostic history, ensuring that previously generated reports are displayed clearly and in an organized manner. In Test Case FT-004, the explainability feature is tested by selecting a report and viewing the corresponding heatmap; the system accurately generates and displays the highlighted retinal regions used in the AI’s decision-making. Test Case FT-005 focuses on report management, confirming that patients can download their diagnostic report without errors.

Test Case FT-006 examines the built-in chatbot, ensuring that it responds with appropriate general guidance when a patient asks non-clinical questions related to their diagnosis or system usage. Test Case FT-007 validates profile management functionality by checking whether changes to the patient’s personal information are correctly applied and updated. Finally, Test Case FT-008 tests the logout process, confirming that the system securely ends the patient session and returns them to the login screen. All test cases passed successfully, demonstrating that the patient workflow operates smoothly and meets the intended system requirements.

6.1.4 Integration Testing

Integration testing is a crucial phase in software development that focuses on combining individual units or modules and testing them as a group. This process helps identify defects in the interactions between modules, ensuring they communicate correctly, and function together as intended. By detecting interface issues, logical errors, and potential data mismatches, integration testing confirms that the application operates smoothly in a real-world environment. It ultimately enhances software reliability before proceeding to further testing stages [10].

Integration Testing 1: Clinician Workflow

Objective: To verify that all clinician-related modules such as login, image upload, AI diagnosis, explainability, and report handling - integrate smoothly and function correctly together.

Table 6.1.4.19: Clinician Workflow

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	Clinician logs in and accesses dashboard	Login Module, Dashboard Module	Clinician successfully logs in and dashboard loads correctly	Pass

IT-002	Clinician uploads an image and initiates diagnosis	Image Upload Module, AI Diagnosis Module, Pre-processing Module	Image is uploaded and diagnosis is generated	Pass
IT-003	Clinician views explainability map	Diagnosis Module, Explainability Module	Explainability heatmap is generated and displayed	Pass
IT-004	Clinician generates and downloads a report	Report Generation Module, File Export Module	Complete diagnostic report is created and downloaded	Pass
IT-005	Clinician updates profile information	Profile Module, Database Module	Profile changes saved and updated immediately	Pass
IT-006	Clinician logs out	Dashboard Module, Logout Module	Clinician is logged out successfully	Pass

The integration testing for the clinician workflow verifies that the major diagnostic and management modules operate cohesively. In Test Case IT-001, the login and dashboard modules work together to authenticate the clinician and load the correct interface. Test Case IT-002 ensures proper coordination between the upload, pre-processing, and AI diagnosis modules; the system successfully accepts the image, processes it, and generates a glaucoma prediction. Test Case IT-003 focuses on explainability, confirming that the diagnosis results integrate properly with the heatmap generation module.

In Test Case IT-004, the system successfully compiles diagnostic results, explainability maps, and patient metadata to generate a complete report that can be downloaded through the file export module. Test Case IT-005 validates the interaction between the profile settings and database modules by updating clinician details and reflecting them in real time. Finally, Test Case IT-006

verifies that the logout module properly ends the session and returns the clinician to the login screen. All tests passed, confirming smooth integration across clinician panels.

Integration Testing 2: Patient Workflow

Objective: To ensure that patient-facing modules such as login, report viewing, chatbot assistance, and profile management work together correctly and provide a smooth experience.

Table 6.1.4.20: Patient Workflow

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	Patient logs in and navigates to dashboard	Login Module, Navigation Module	Patient logs in and is redirected to the correct dashboard	Pass
IT-002	Patient views diagnostic history	Reports Module, Database Module	Previous reports are retrieved and displayed correctly	Pass
IT-003	Patient opens explainability map from a report	Reports Module, Explainability Viewer Module	Explainability heatmap loads with accurate visual overlay	Pass
IT-004	Patient downloads a previous report	Reports Module, File Export Module	Report downloads successfully	Pass
IT-005	Patient interacts with chatbot for guidance	Chatbot Module, Knowledge Base	Chatbot responds correctly with informational guidance	Pass
IT-006	Patient updates profile information	Profile Module, Database Module	Profile updates saved and shown immediately	Pass

IT-007	Patient logs out	Dashboard Module, Logout Module	Patient logged out and returned to login screen	Pass
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Integration testing for the patient workflow ensures that all patient-oriented modules interact smoothly and provide a consistent experience. Test Case IT-001 confirms that the login and navigation modules work together to authenticate the patient and deliver them to the correct interface. Test Case IT-002 validates that the reports module correctly retrieves stored reports from the database and displays them without errors.

In Test Case IT-003, the explainability viewer integrates properly with the stored diagnostic data, allowing patients to view heatmaps that highlight important retinal regions. Test Case IT-004 checks that the report download process works seamlessly across the reports and file export modules. Test Case IT-005 verifies the chatbot integration, confirming that system queries are processed and answered as expected. Test Case IT-006 confirms smooth communication between the profile and database modules during updates. Finally, Test Case IT-007 ensures that the logout module properly ends the session and redirects the patient to the login screen. All integration tests passed, confirming that the patient workflow components are well-connected and functioning as intended.

6.2 Tools and Technologies

Table 6.2.21: Tools and Technologies for Proposed Projects

Tools And Technologies	Tools	Version	Rationale
	Visual Studio Code	1.106	IDE used for development
	Jupyter Notebook	6.2.0	Development and testing of models
	MySQL Workbench	8.0.44	DB, Storage
	Figma	-	UI/UX Design and Mockup Creation
	MS PowerPoint	2013	Presentation Creation
	MS Word	2013	Project Documentation
	Git / GitHub	2.30	Version control and collaboration
	Technology	Version	Rationale
	Python	3.11	Programming language for Backend
	OpenCV	4.5.2	Image processing
	TensorFlow	2.6.0	Machine learning model development

	Keras	2.4.3	High-level neural network API
	Scikit-learn	0.24.2	Machine learning
	Grad-CAM	N/A	Explainable AI techniques
	FastAPI	0.122.0	Web framework for backend services
	React.js / Next.js	19.2.0	Modern frontend frameworks
	MySQL	8.4.6	Database management and storage

6.3 Summary

This chapter presented the different testing approaches applied to the GlaucoVision system, including unit testing, functional testing, and integration testing. Each core module such as user authentication, fundus image upload, AI-based diagnosis, explainability generation, and report handling was carefully tested using both valid and edge-case inputs to verify correctness and reliability. The test cases were executed to evaluate how the system behaves under normal and unexpected conditions, ensuring that each feature meets its intended functionality.

Expected outcomes were compared against actual system responses, and each test case was documented with a Pass or Fail status based on the system's performance. Overall, the testing process confirmed that GlaucoVision functions effectively, providing a stable and accurate experience for both clinicians and patients.

Chapter 7

Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The GlaucoVision system has been developed to support clinicians and patients by providing an easier, faster, and more organized way to detect glaucoma using fundus images. The main goal of this project was to create a digital platform where users can upload eye images, receive AI-generated predictions, view explainability reports, and keep all their diagnostic information in one place. The system also helps maintain patient histories, making it simpler for both patients and clinicians to track changes over time.

Before GlaucoVision, users had to rely on traditional manual processes, such as physically storing reports, depending on expert interpretation, and dealing with scattered medical information. Now, with GlaucoVision, this entire process has become more structured, user-friendly, and accessible. All essential features—including uploading images, viewing diagnosis results, generating explainable visual maps, and downloading reports—are available through a single unified platform.

Another major advantage of the system is its clean and easy-to-use interface. Both clinicians and patients can use the platform without needing any technical training. Whether someone wants to check their analysis results, view previous scans, or generate new reports, the system provides a smooth and simple experience.

Throughout the development process, the system was thoroughly tested to ensure quality. Features like user authentication, image upload, AI prediction, explainability visualization, dashboard navigation, and report downloads were checked to make sure they work correctly and reliably.

In conclusion, GlaucoVision successfully achieves its purpose by simplifying glaucoma detection, improving accessibility to diagnostic information, and making the clinical workflow more organized. It is a step forward in bringing intelligent, tech-friendly solutions to eye care and helping users stay informed about their eye health.

7.2 Future Work

Although the current version of GlaucoVision performs well and meets its main objectives, there are several improvements and new features planned for future updates. These enhancements will help make the system more powerful, advanced, and user-centered.

7.2.1 Integration of Advanced AI Models

Future versions can include more powerful deep learning models that offer higher accuracy, better sensitivity for early-stage glaucoma, and improved explainability.

7.2.2 Mobile Application Version

Developing a mobile app for both Android and iOS will allow users to upload images and view reports directly from their phones, making the system more convenient and accessible.

7.2.3 Role-Based Features and Notifications

Introduce role-specific notifications, reminders for follow-up scans, and alerts for new reports, allowing users to stay updated at all times.

7.2.4 Enhanced Security Features

Strengthen system security by adding features such as two-factor authentication (2FA) and improved encryption to protect medical information.

7.2.5 Strong Security Measure

Improve data protection by adding advanced login features like two-factor authentication (2FA) and better encryption.

7.2.6 Performance Optimization

Improve processing time, speed of AI predictions, and dashboard responsiveness so the system can handle larger datasets and more users efficiently.

7.2.7 Integration with Clinic Systems

In the future, GlaucoVision can be connected with hospital information systems to automate record updates and ensure smoother clinical workflows.

By continuing to upgrade and refine GlaucoVision, the system will remain useful, modern, and valuable for both patients and clinicians. These improvements will help it adapt to future needs and provide a better healthcare experience for everyone.

8 References

- [1] Abràmoff, M. D., Lavin, P. T., Birch, M., Shah, N., & Folk, J. C. (2018). Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *NPJ Digital Medicine*, 1(1), 1–8.
- [2] Chen, X., Xu, Y., Wong, D. W. K., Wong, T. Y., & Liu, J. (2015). Glaucoma detection based on deep convolutional neural network. *Engineering Applications of Artificial Intelligence*, 45, 26–33.
- [3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- [4] A. N. a. B. Meyer, "Seamless Requirements," *arXiv preprint arXiv:1704.04189*, 2017.
- [5] Langlotz, C. P. (2019). Will artificial intelligence replace radiologists? *Radiology*, 290(2), 365–366.
- [6] Lin, J. W., & Wang, Y. (2019). Explainable AI and interpretability in deep learning medical imaging. *Journal of Medical Artificial Intelligence*, 2(1), 1–12.
- [7] Paszke, A., Gross, S., Massa, F., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32, 8024–8035.
- [8] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144.
- [9] D. G. L. J. Fewster, "Software Test Automation: Effective Use of Test Execution Tools," *Addison-Wesley Professional*, pp. 1-250, 1999.
- [10] C. Beizer, "Software Testing Techniques," *Van Nostrand Reinhold*, pp. 45-67, 1990.